



(54) **DISCHARGE-FREE READ OPERATIONS FOR HIGH BANDWIDTH NONVOLATILE MEMORY DEVICES**

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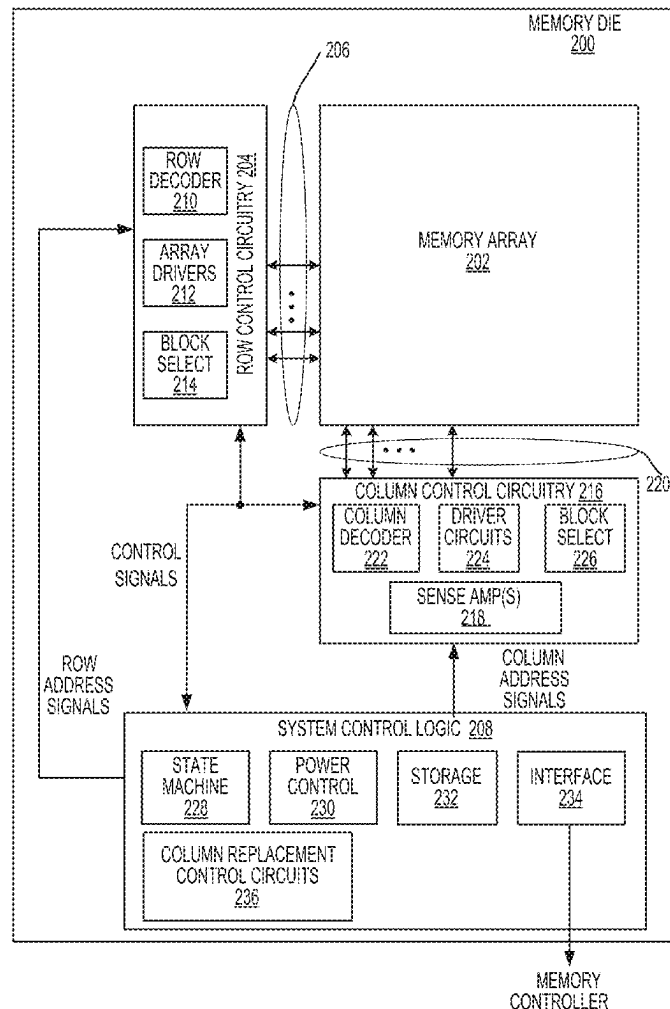
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(57) **ABSTRACT**

The memory device includes a memory block with a plurality of strings with non-volatile memory cells, each of which is coupled to one of a plurality of word lines. The memory device also includes control circuitry that is coupled to the memory array. The control circuitry is configured to perform a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage. The control circuitry is also configured to perform a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage. The unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.



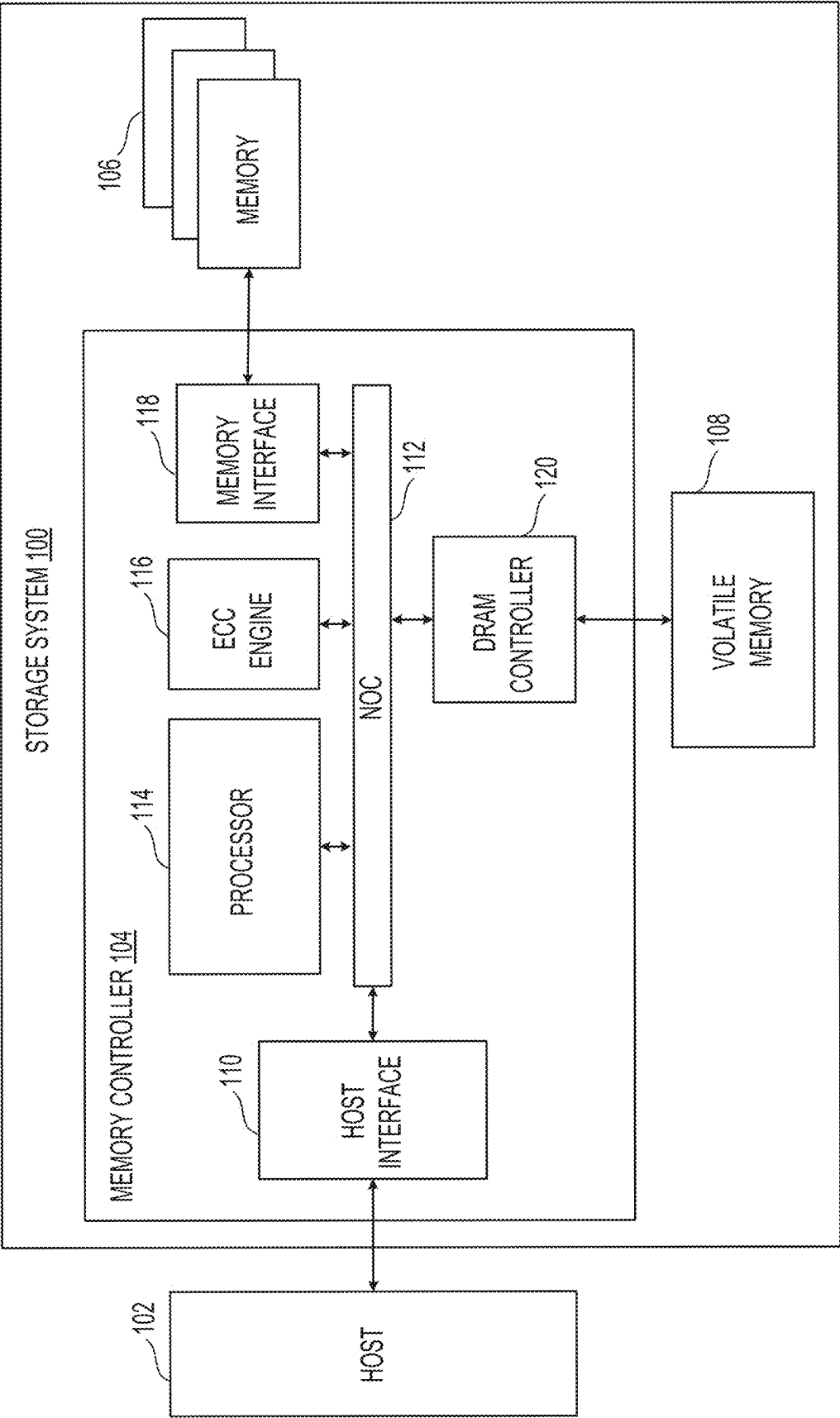
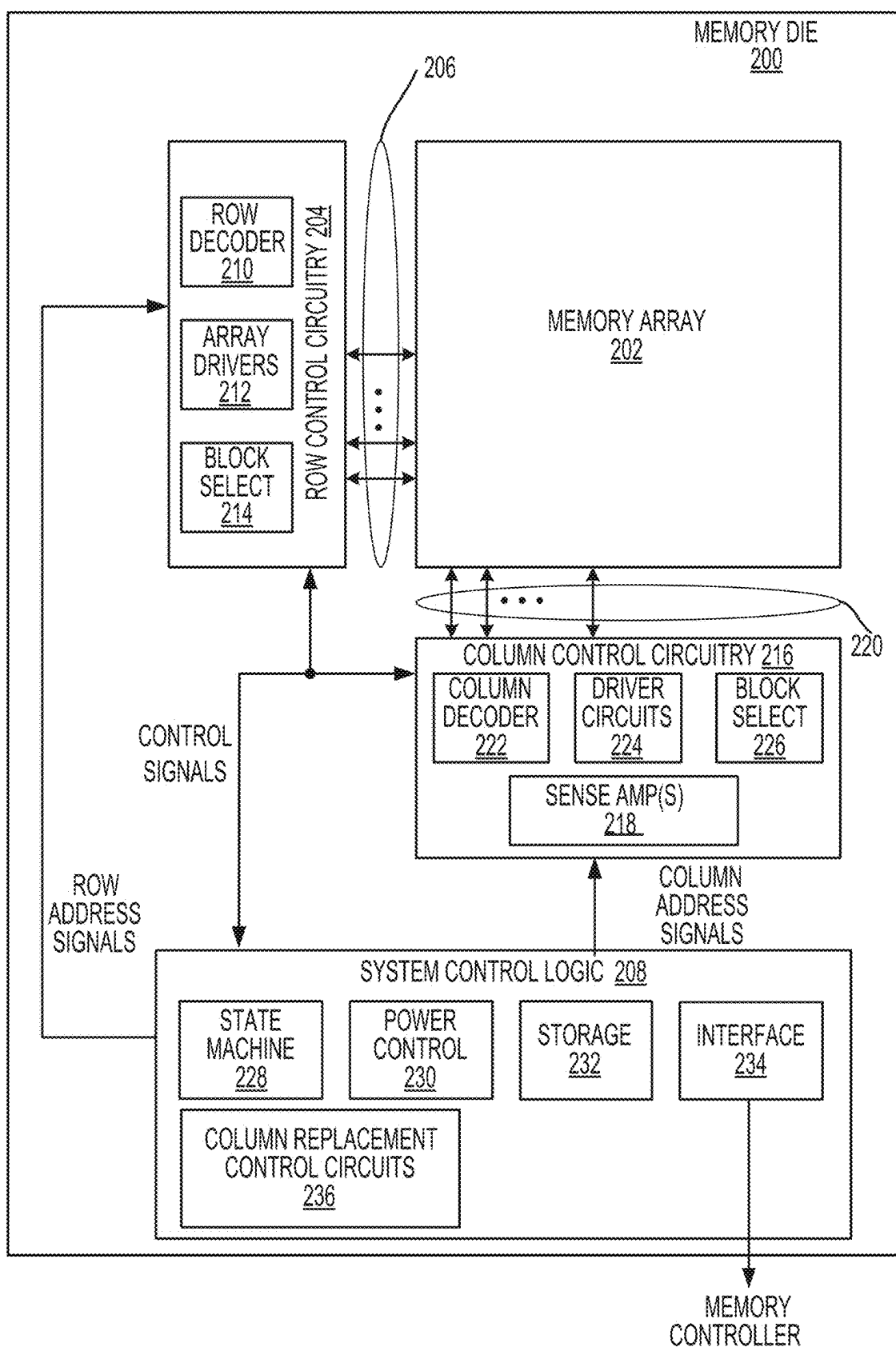


FIG. 1



**FIG. 2A**

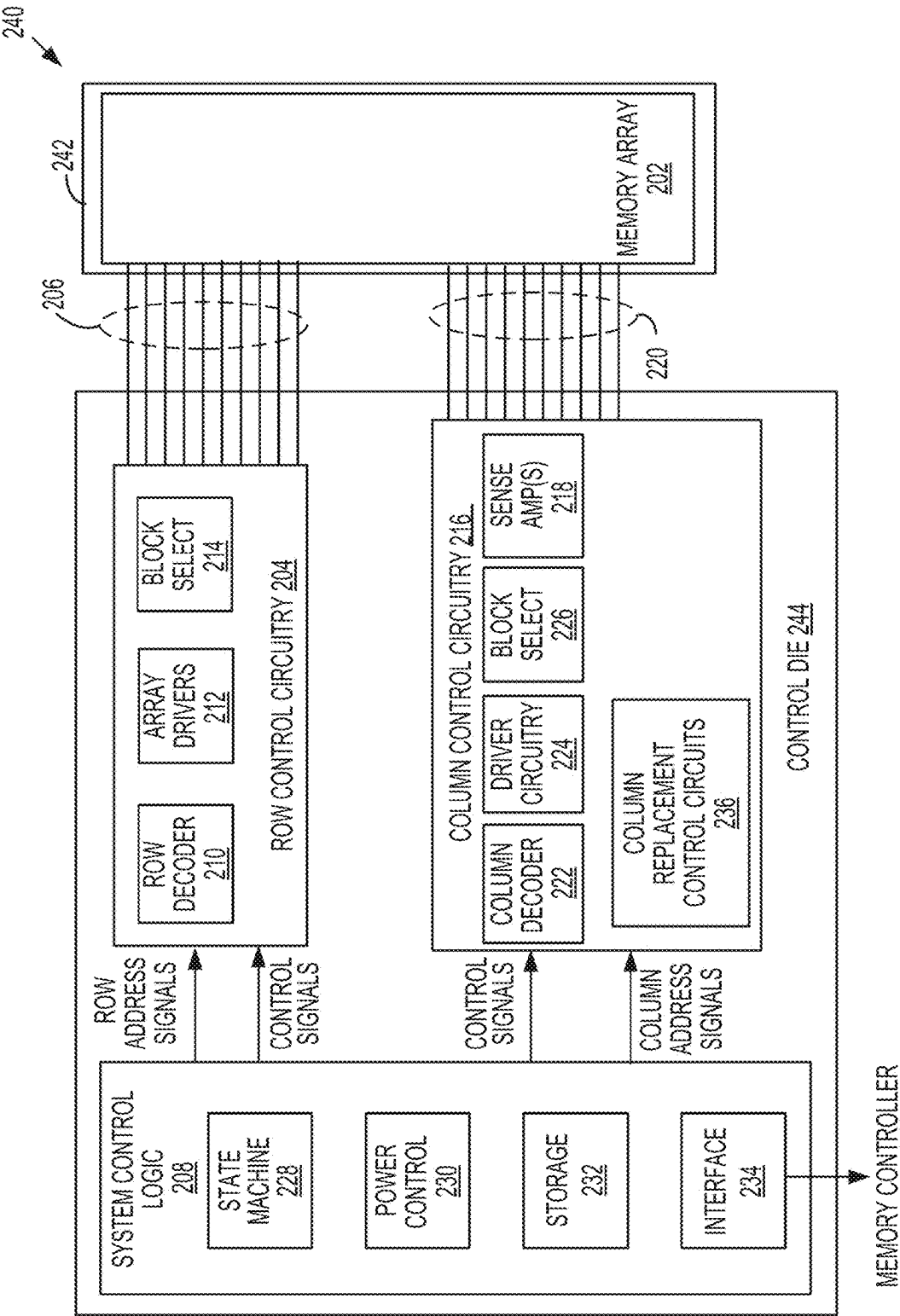
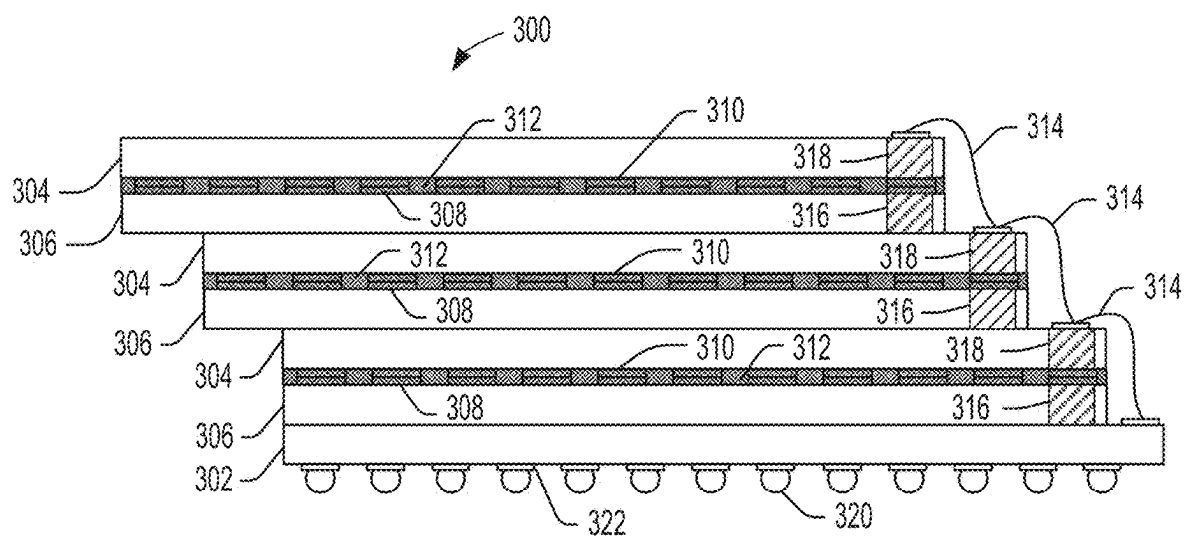
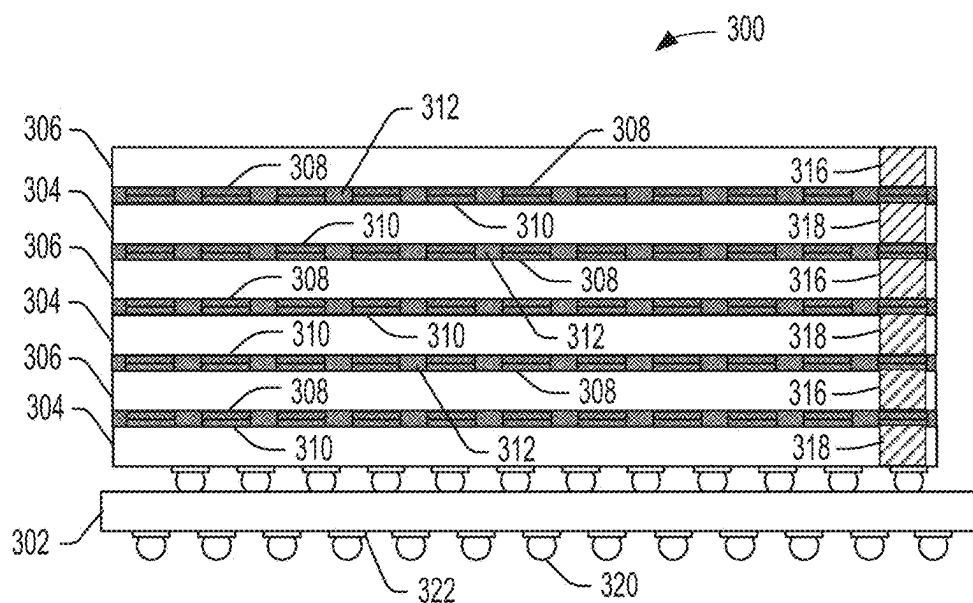


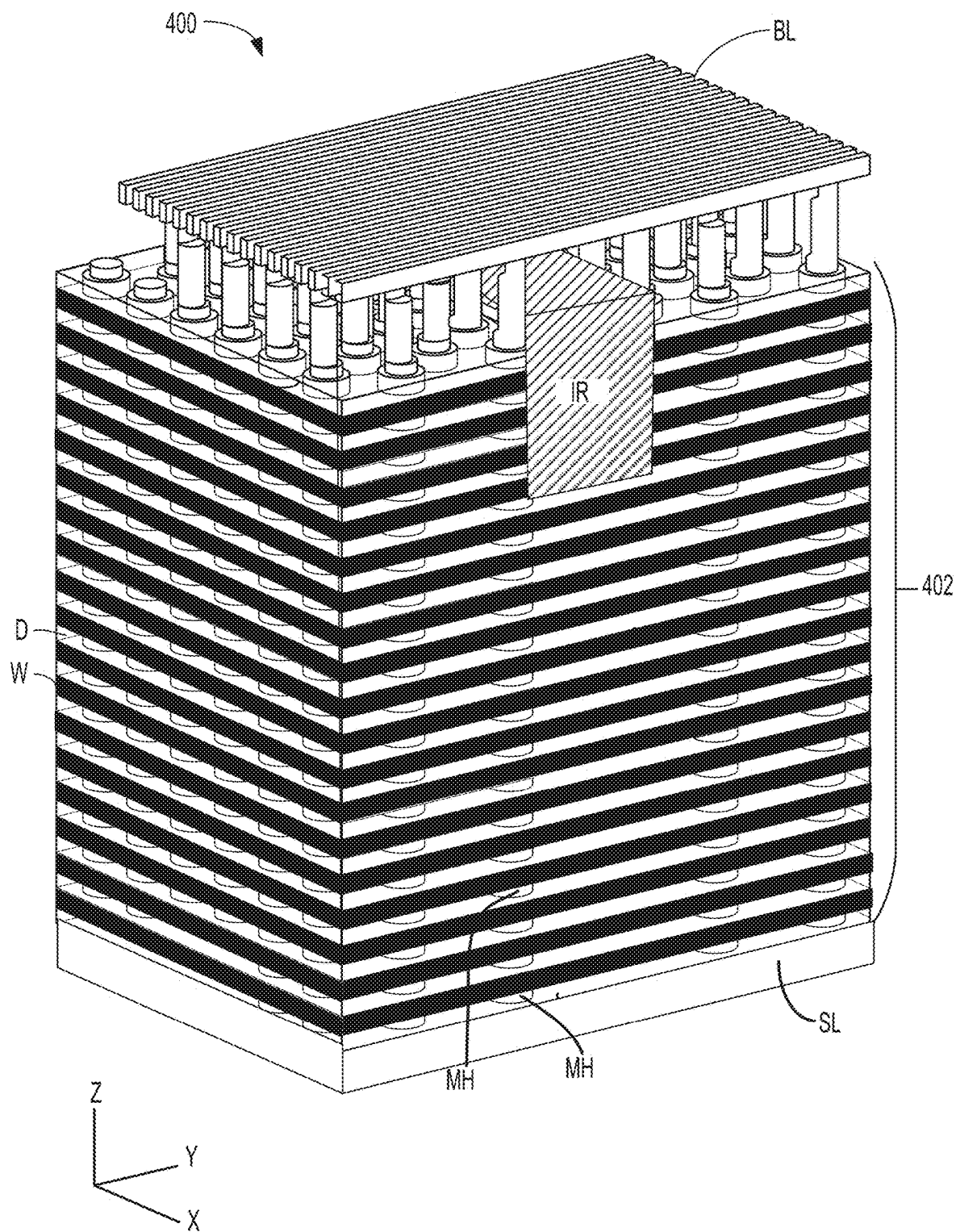
FIG. 2B



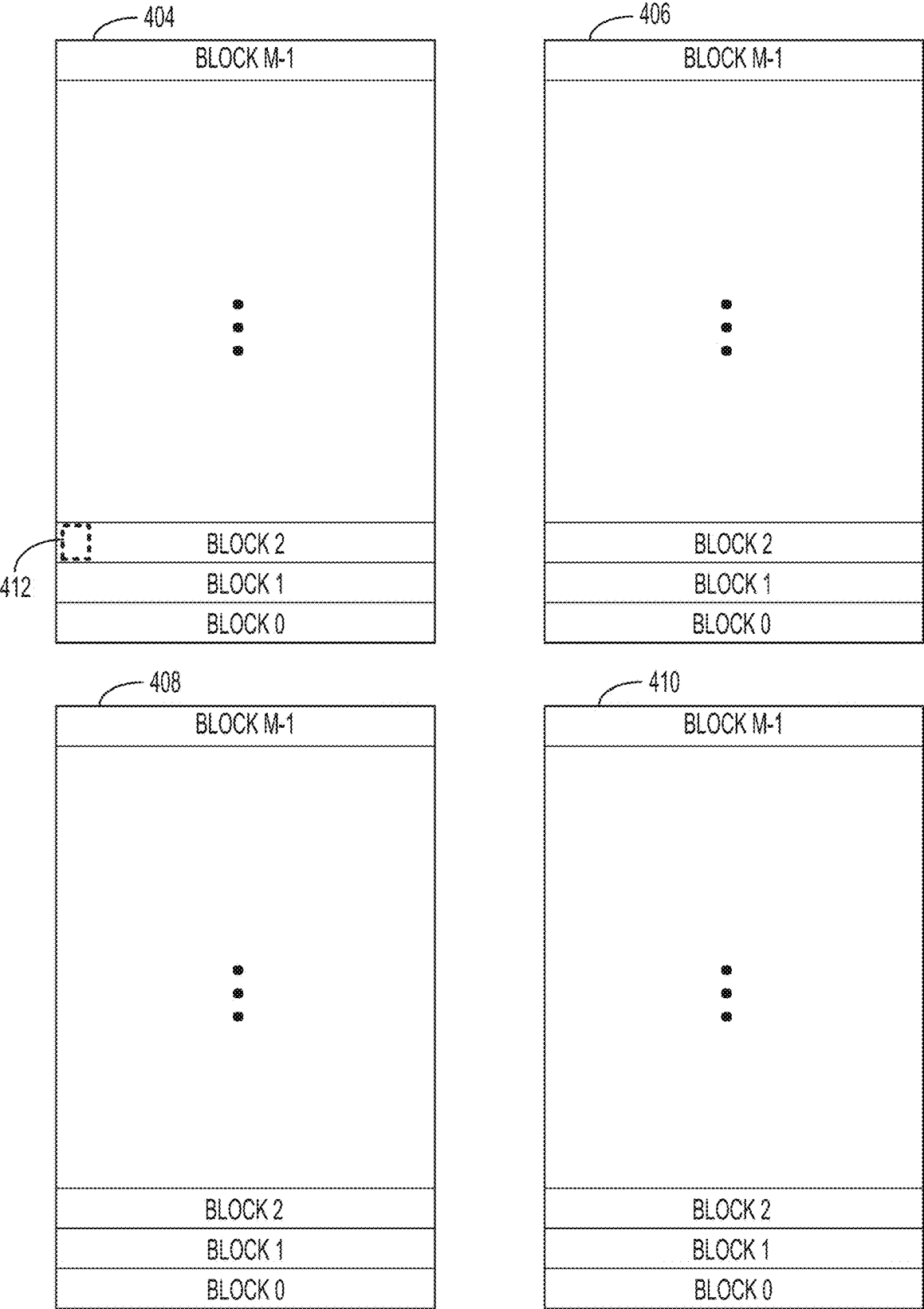
**FIG. 3A**



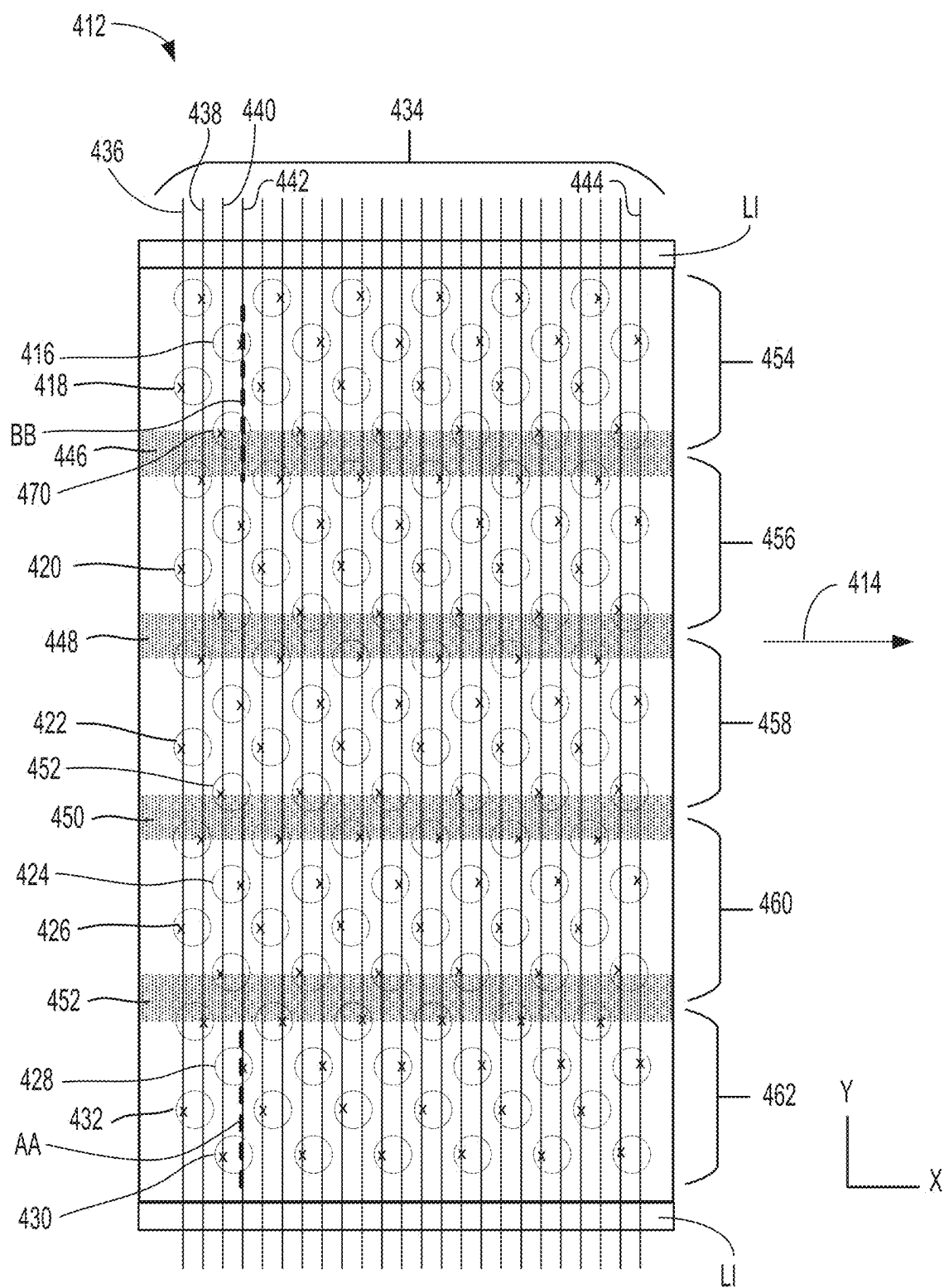
**FIG. 3B**



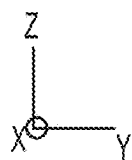
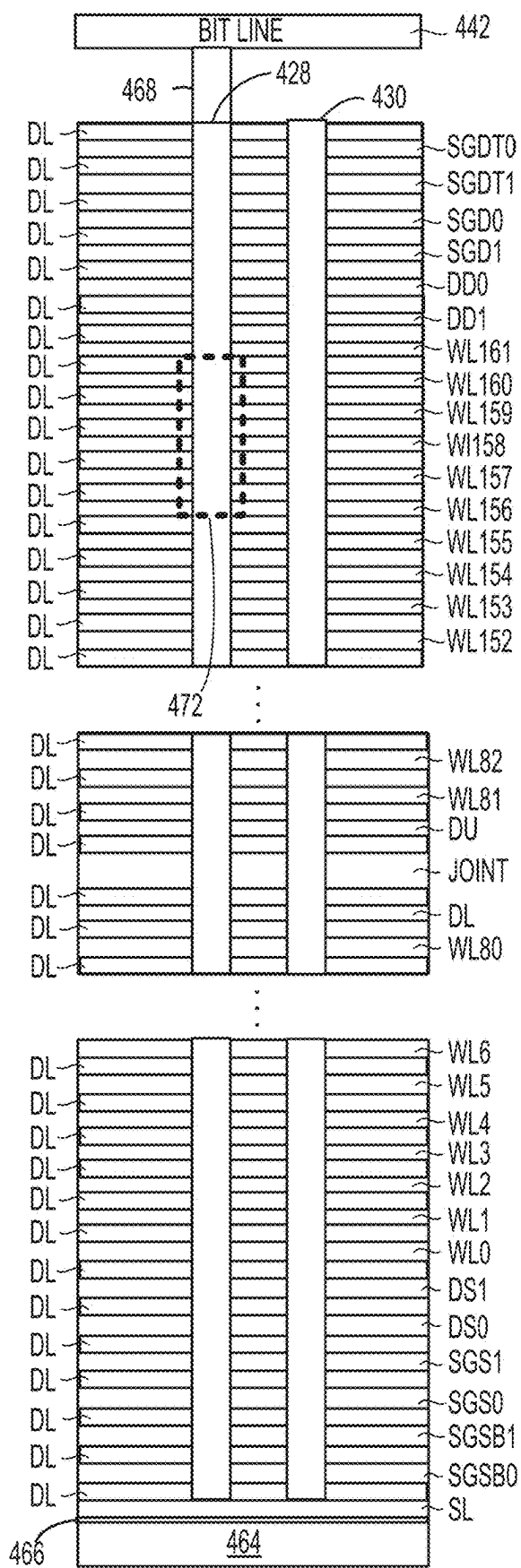
**FIG. 4A**



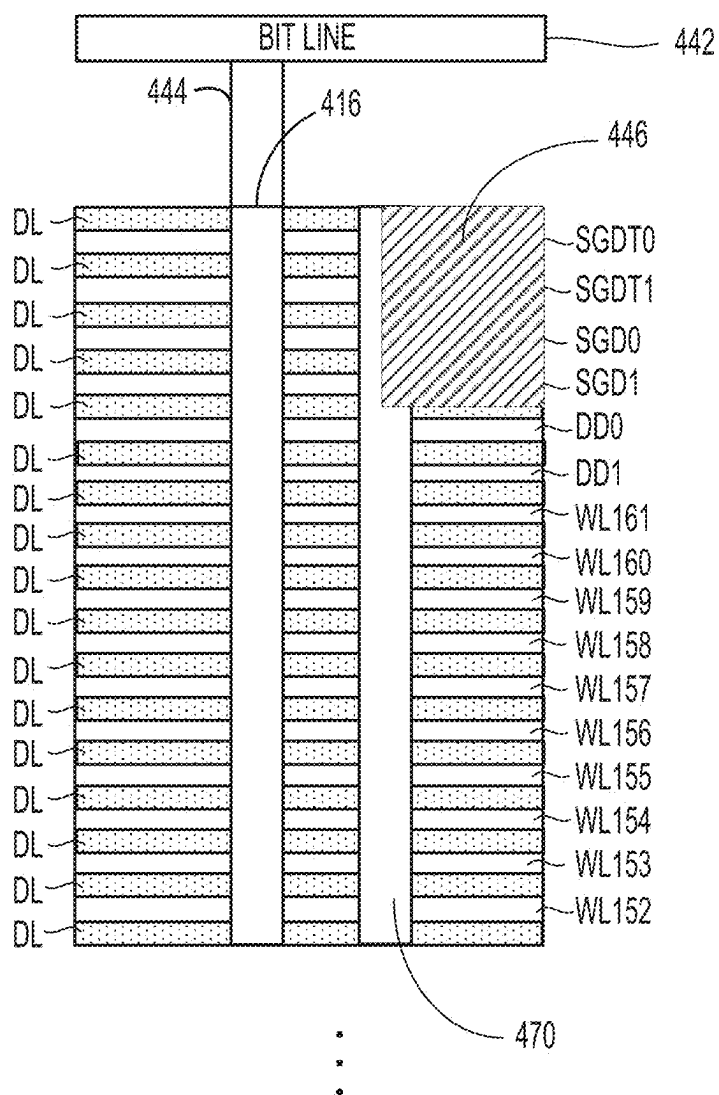
**FIG. 4B**



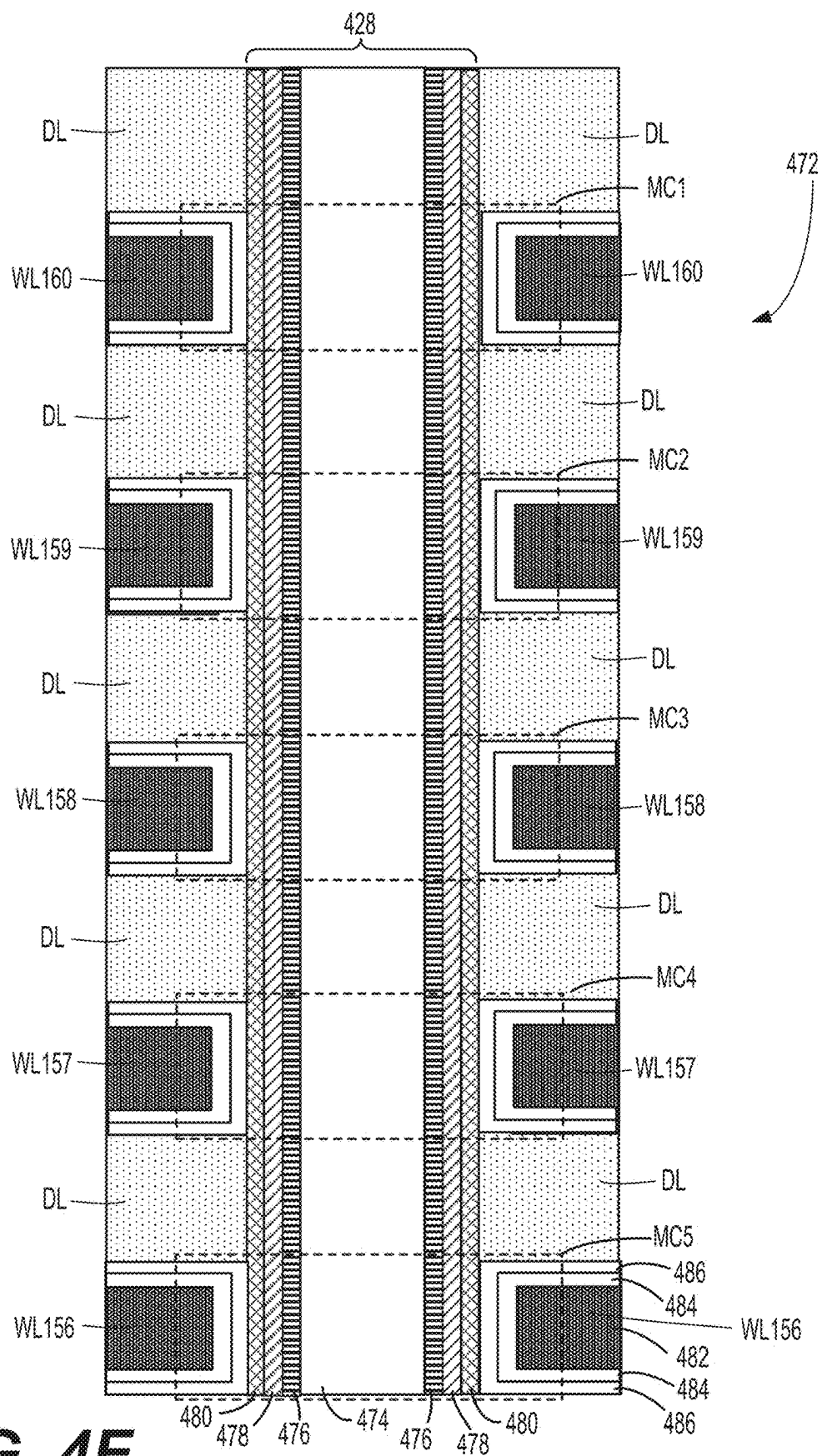
**FIG. 4C**



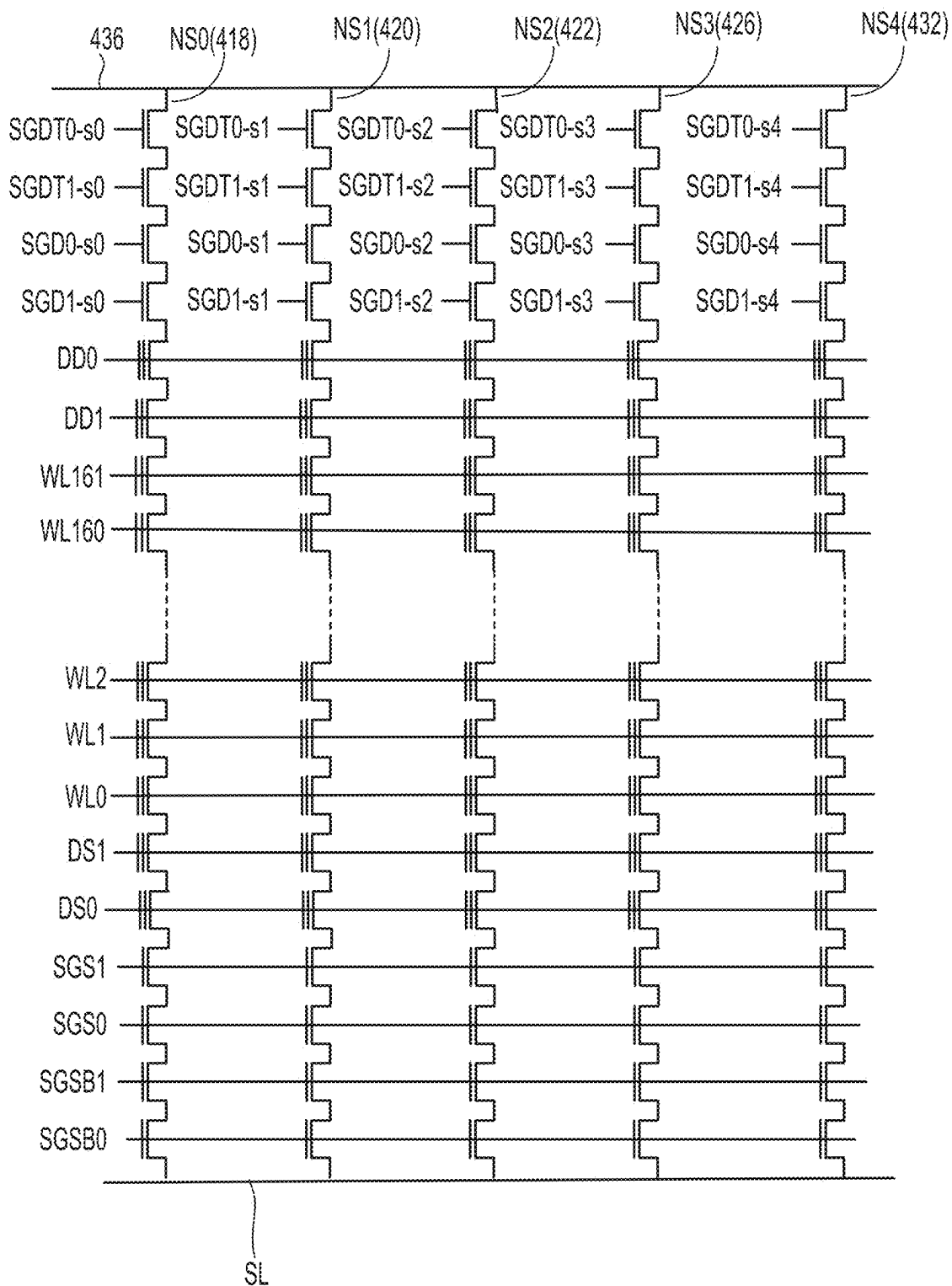
**FIG. 4D**



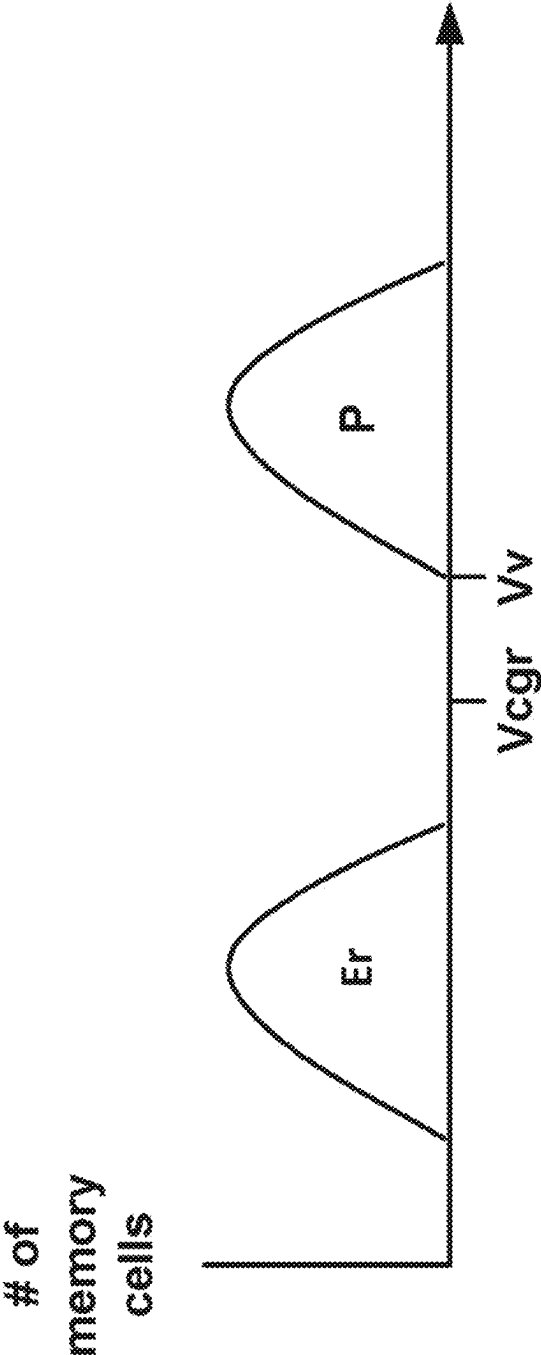
**FIG. 4E**



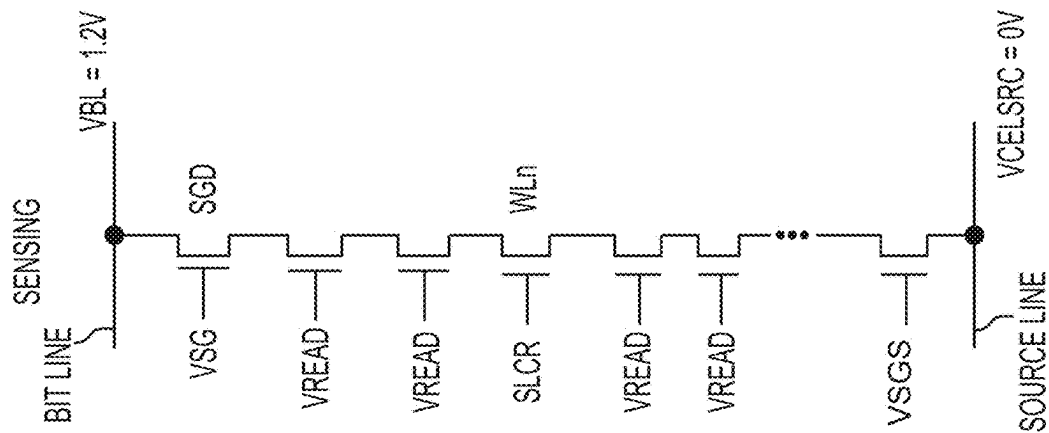
**FIG. 4F**



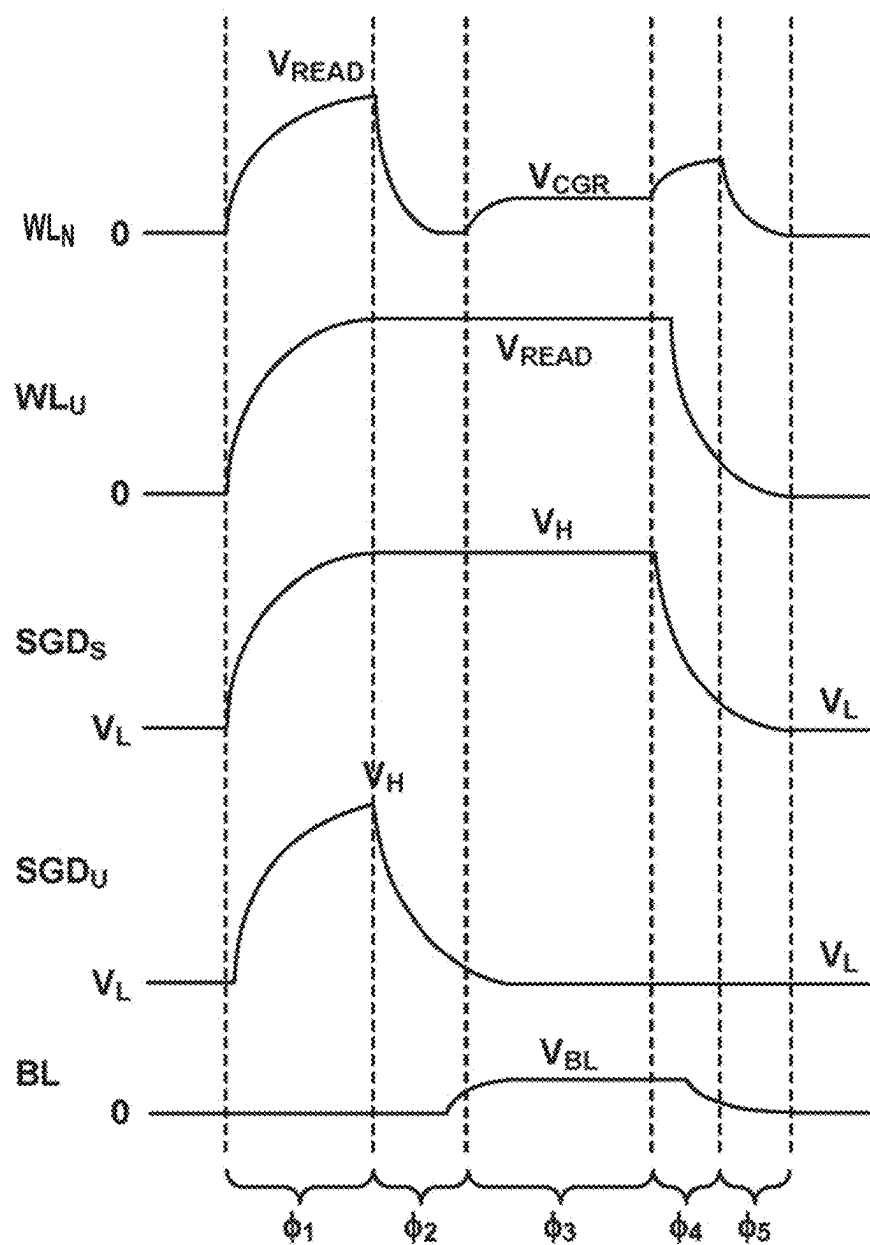
**FIG. 4G**



**FIG. 5**

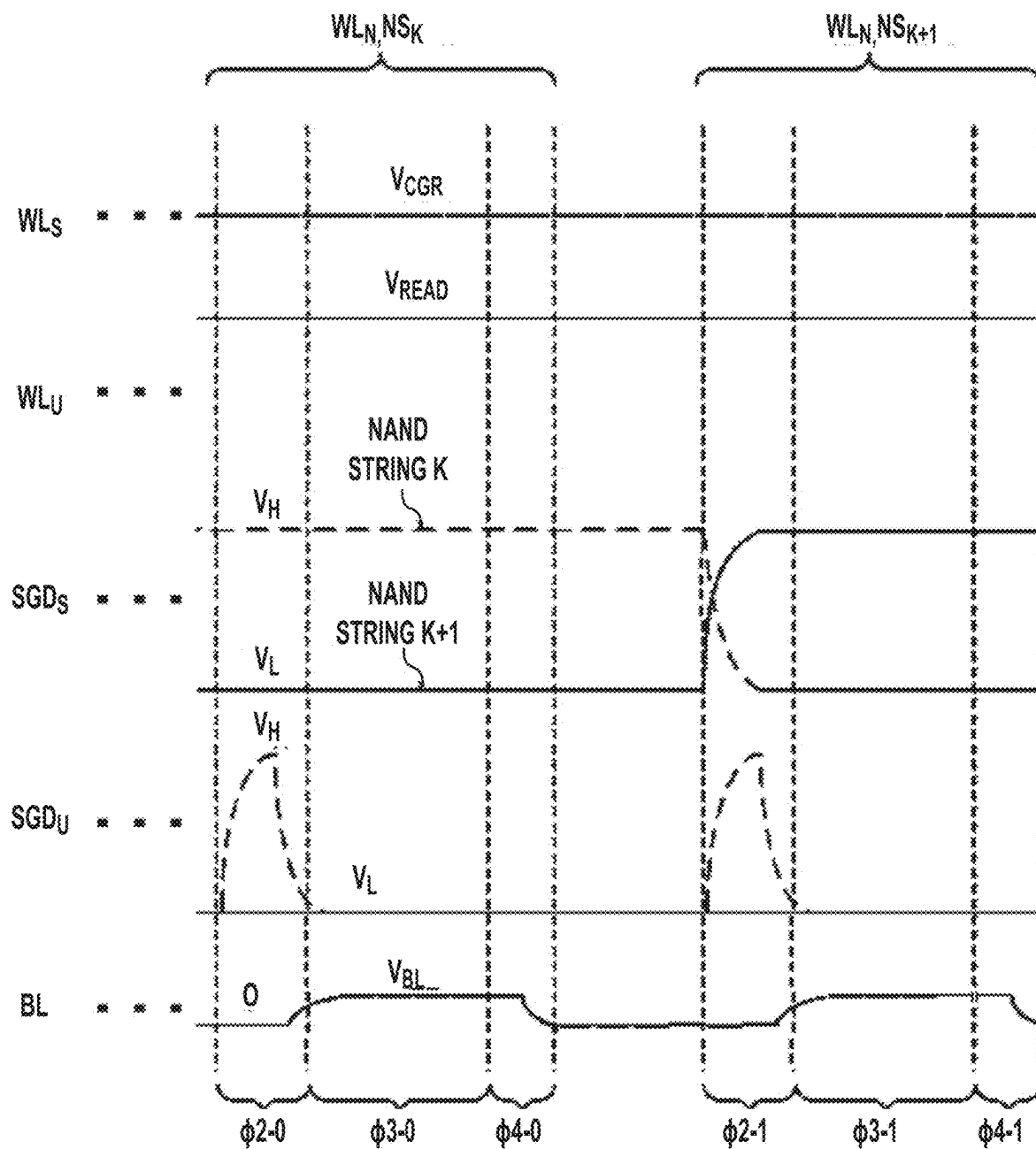


**FIG. 6**

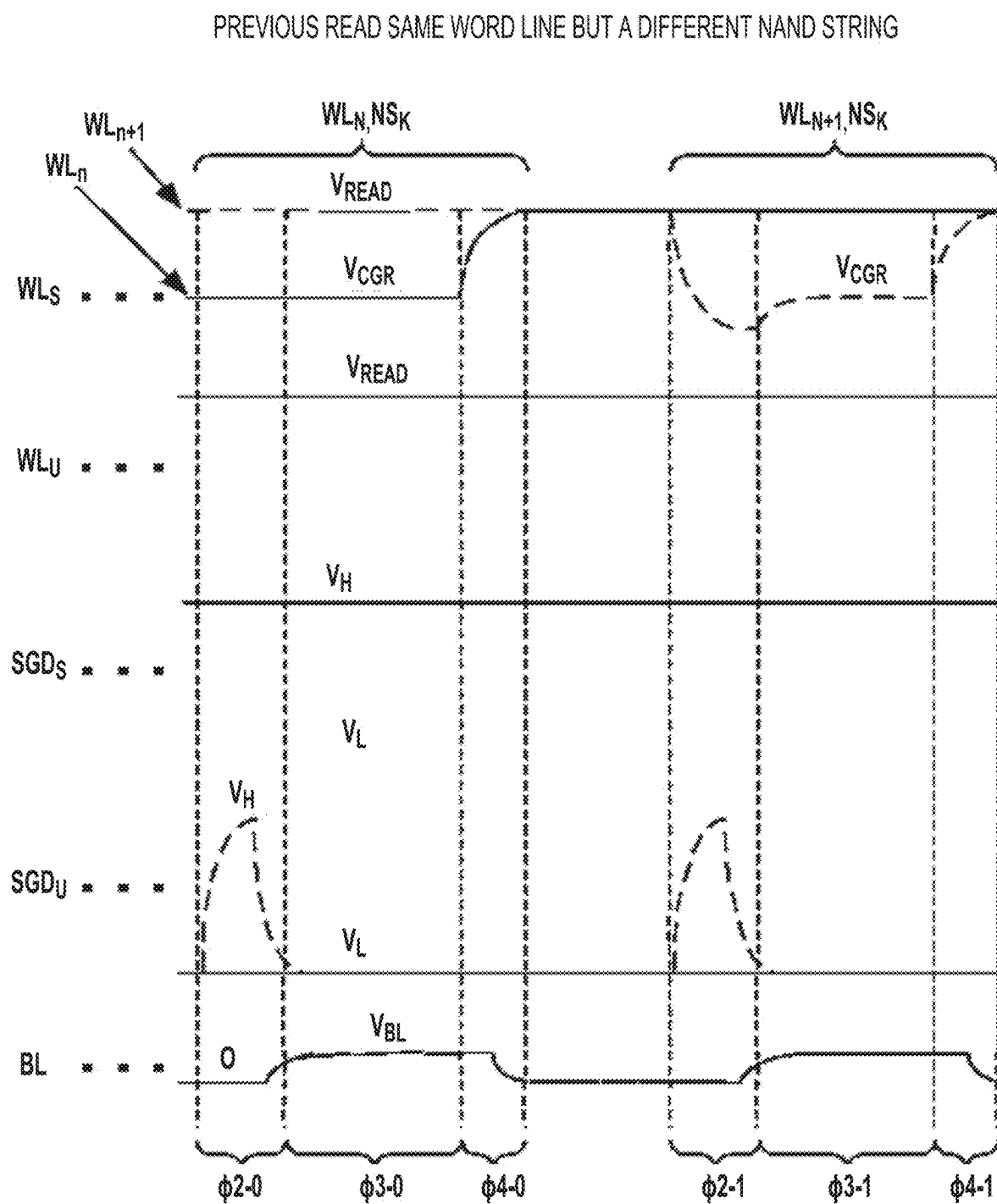


**FIG. 7**

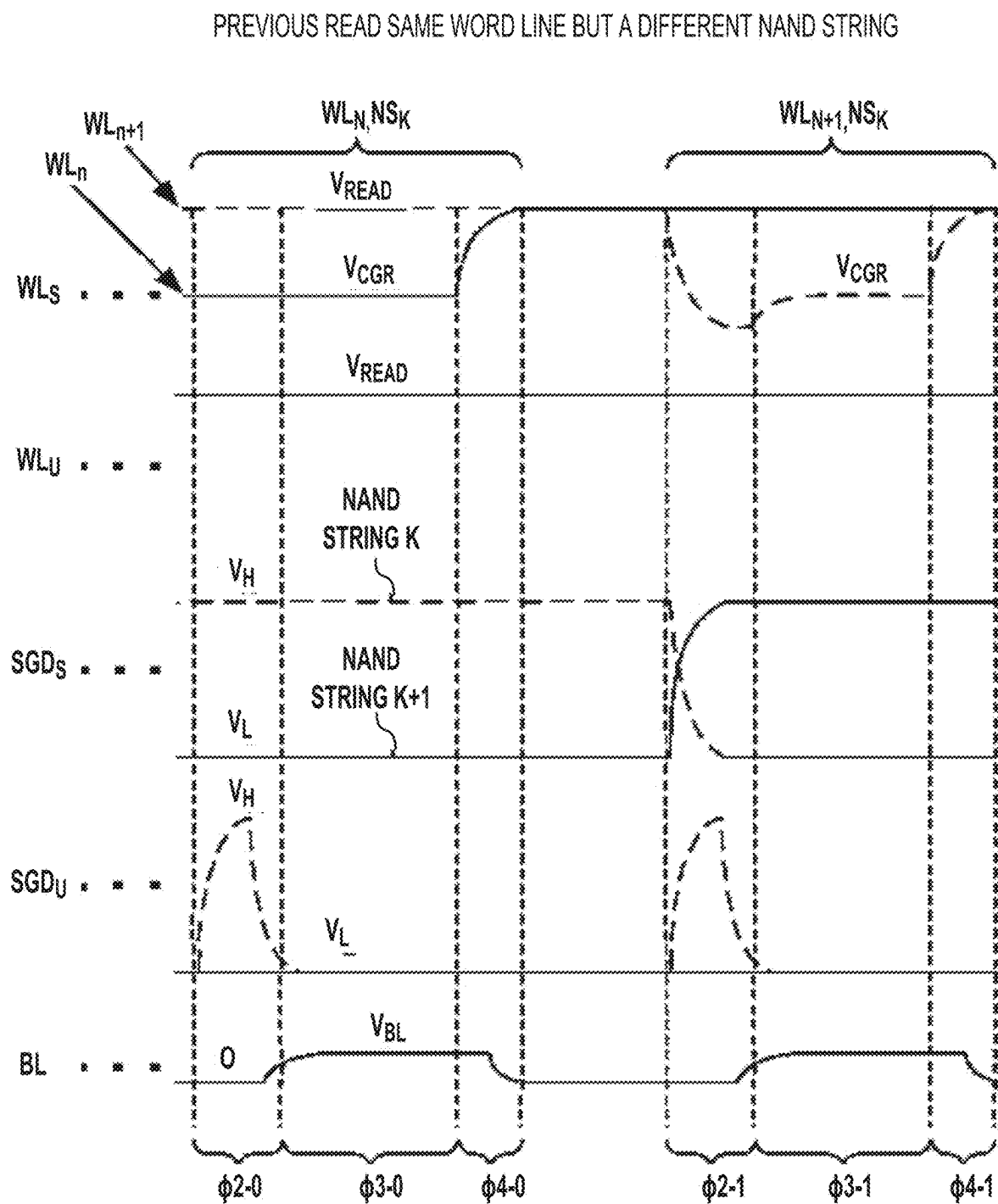
PREVIOUS READ SAME WORD LINE BUT A DIFFERENT NAND STRING



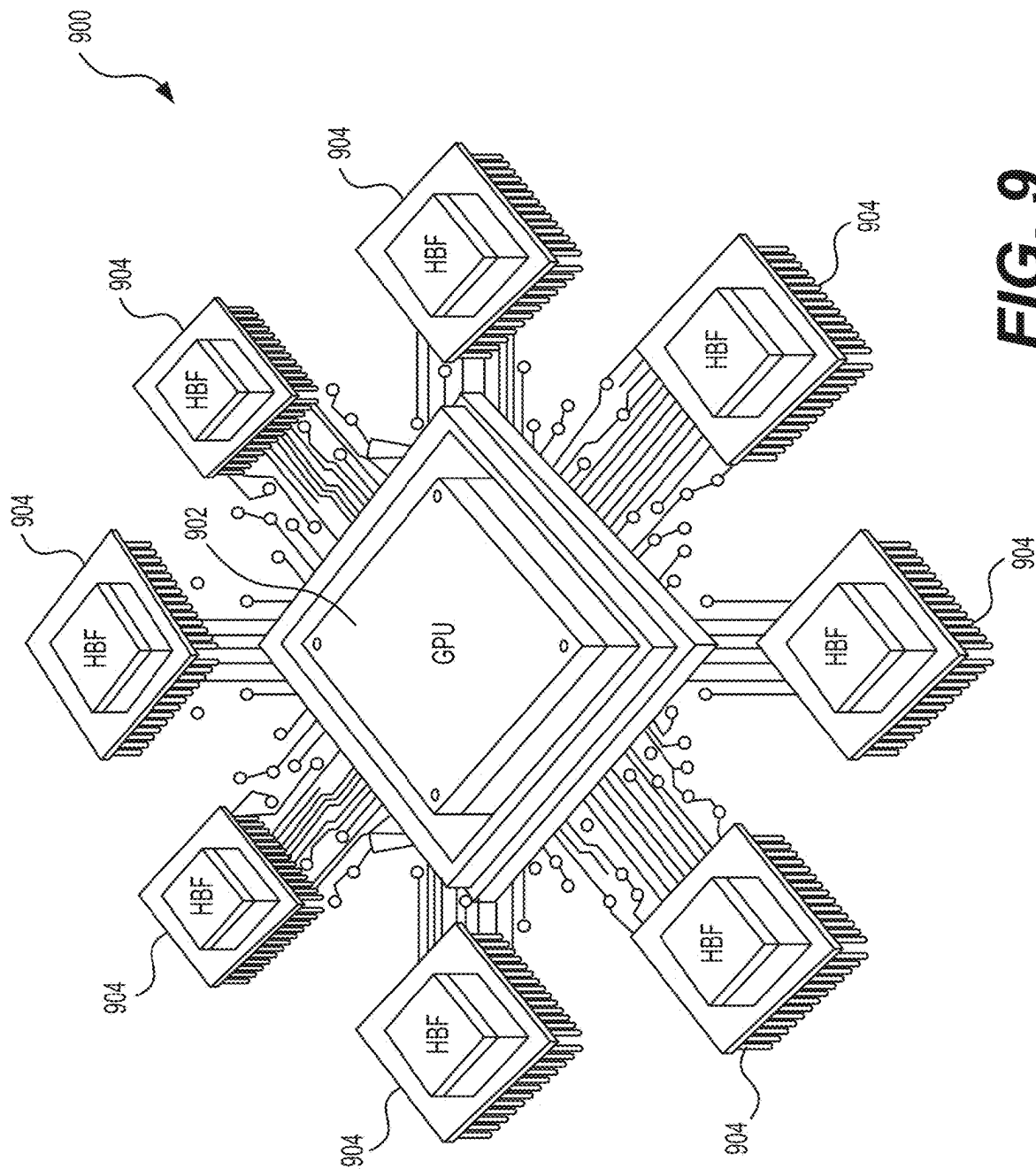
**FIG. 8A**



**FIG. 8B**



**FIG. 8C**



**FIG. 9**

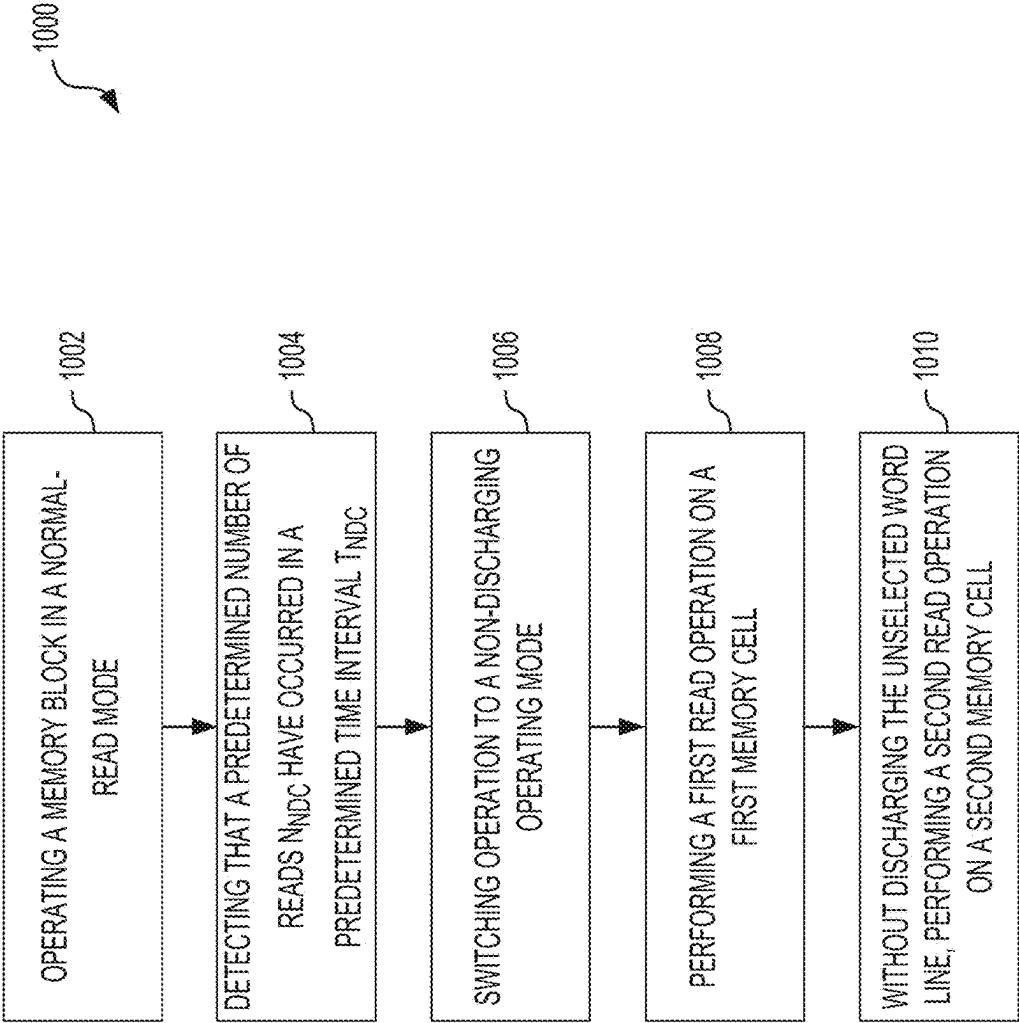


FIG. 10

## DISCHARGE-FREE READ OPERATIONS FOR HIGH BANDWIDTH NONVOLATILE MEMORY DEVICES

### CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of U.S. Provisional App. No. 63/561,196, filed Mar. 4, 2024, entitled “DISCHARGE-FREE READ OPERATIONS FOR HIGH BANDWIDTH NONVOLATILE MEMORY DEVICE,” the entire contents of which is herein incorporated by reference.

### BACKGROUND

#### 1. Field

[0002] The present disclosure is related generally to non-volatile memory and, more particularly to improved memory devices that are optimized to operate at very high read performance and with a very low power consumption.

#### 2. Related Art

[0003] Semiconductor memory is widely used in various electronic devices such as cellular telephones, digital cameras, personal digital assistants, medical electronics, mobile computing devices, servers, solid state drives, non-mobile computing devices and other devices. Semiconductor memory may be non-volatile memory or volatile memory. A non-volatile memory allows information to be stored and retained even when the non-volatile memory is not connected to a source of power (e.g., a battery).

[0004] Non-volatile memory devices include one or more memory chips having multiple arrays of memory cells. The memory arrays may have associated decoders and circuits for performing read, write, and erase operations. Memory cells within the arrays may be arranged in horizontal rows and vertical columns. Each row may be addressed by a word line, and each column may be addressed by a bit line. Data may be loaded into columns of the array using a series of data busses. Each column may hold a predefined unit of data, for instance, a word encompassing two bytes of information.

[0005] In some applications, semiconductor memory is used to store very large amounts of data that are repeatedly accessed (e.g., read) very rapidly. For example, in some machine learning applications, large language models that include a terabyte (or more) of data must be stored in memory and retrieved at a very high data rate. Accordingly, such applications require very high bandwidth and low power.

[0006] Currently, high bandwidth volatile memory devices (e.g., DRAM memory devices called “high bandwidth memory” or “HBM”) are used for such applications. Non-volatile memory (e.g., NAND) is significantly less expensive than DRAM, but the bandwidth of conventional NAND memory devices is too low, and the power consumption of conventional NAND memory devices is too high to provide a viable alternative to HBM devices. Therefore, there is a need to provide high bandwidth, low power non-volatile memory.

### SUMMARY

[0007] One aspect of the present disclosure is related to a memory device that includes a memory block with a plu-

rality of strings. Each string comprises a plurality of non-volatile memory cells, and each non-volatile memory cell is coupled to one of a plurality of word lines. The memory device also includes control circuitry that is coupled to the memory array. The control circuitry is configured to perform a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage. The control circuitry is also configured to perform a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage. The unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.

[0008] According to another aspect of the present disclosure, the first non-volatile memory cell is in the same word line as the second non-volatile memory cell and is in a different string than the second non-volatile memory cell.

[0009] According to yet another aspect of the present disclosure, the first non-volatile memory cell is in the same string as the second non-volatile memory cells and is in a different word line from the second non-volatile memory cell.

[0010] According to still another aspect of the present disclosure, the first non-volatile memory cell is in a different word line than and is in a different string than the second non-volatile memory cell.

[0011] According to a further aspect of the present disclosure, the first and second memory cells are programmed to one bit of data per memory cell.

[0012] According to yet a further aspect of the present disclosure, during the first read operation, the control circuitry discharges a current through the first non-volatile memory cell. During the second read operation, the control circuitry discharges a current through the second non-volatile memory cell.

[0013] Another aspect of the present disclosure is related to a method of operating a memory device. The method includes the step of preparing a memory block that includes a plurality of strings. Each string has a plurality of non-volatile memory cells, and each non-volatile memory cell is coupled to one of a plurality of word lines. The method proceeds with the step of performing a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage. The method continues with the step of performing a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage. The unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.

[0014] According to another aspect of the present disclosure, the first non-volatile memory cell is in the same word line as the second non-volatile memory cell and is in a different string than the second non-volatile memory cell.

[0015] According to yet another aspect of the present disclosure, the first non-volatile memory cell is in the same string as the second non-volatile memory cells and is in a different word line from the second non-volatile memory cell.

[0016] According to still another aspect of the present disclosure, the first non-volatile memory cell is in a different word line than and is in a different string than the second non-volatile memory cell.

[0017] According to a further aspect of the present disclosure, the first and second memory cells are programmed to one bit of data per memory cell.

[0018] According to yet a further aspect of the present disclosure, during the first read operation, the control circuit discharges a current through the first non-volatile memory cell. During the second read operation, the control circuit discharges a current through the second non-volatile memory cell.

[0019] Yet another aspect of the present disclosure is related to a computing system that includes a processing unit and a plurality of high bandwidth flash packages in electrical communication with the processing unit. At least one of the high bandwidth flash packages includes a memory block with a plurality of strings. Each string comprises a plurality of non-volatile memory cells, and each non-volatile memory cell is coupled to one of a plurality of word lines. Each of the high bandwidth flash packages also includes control circuitry that is coupled to the memory array. The control circuit is configured to perform a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage. The control circuitry is also configured to perform a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage. The unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.

[0020] According to another aspect of the present disclosure, the first non-volatile memory cell is in the same word line as the second non-volatile memory cell and is in a different string than the second non-volatile memory cell.

[0021] According to yet another aspect of the present disclosure, the first non-volatile memory cell is in the same string as the second non-volatile memory cells and is in a different word line from the second non-volatile memory cell.

[0022] According to still another aspect of the present disclosure, the first non-volatile memory cell is in a different word line than and is in a different string than the second non-volatile memory cell.

[0023] According to a further aspect of the present disclosure, the first and second memory cells are programmed to one bit of data per memory cell.

[0024] According to yet a further aspect of the present disclosure, during the first read operation, the control circuit discharges a current through the first non-volatile memory cell. During the second read operation, the control circuit discharges a current through the second non-volatile memory cell.

[0025] According to still a further aspect of the present disclosure, the first and second read operations are large language model operations.

[0026] According to another aspect of the present disclosure, the plurality of high bandwidth flash packages are able to communicate with the processing unit during read operations at a bandwidth of at least 3 TB/s.

## BRIEF DESCRIPTION OF THE DRAWINGS

[0027] These and other features and advantages of the subject disclosure will become more readily appreciated when considered in connection with the following description of the presently preferred embodiments, appended claims and accompanying drawings, in which:

[0028] FIG. 1 is a block diagram depicting one embodiment of a storage system;

[0029] FIG. 2A is a block diagram of one embodiment of a memory die;

[0030] FIG. 2B is a block diagram of one embodiment of an integrated memory assembly;

[0031] FIGS. 3A and 3B depict different embodiments of integrated memory assemblies;

[0032] FIG. 4A is a perspective view of a portion of one embodiment of a monolithic three dimensional memory structure;

[0033] FIG. 4B is a block diagram of one embodiment of a memory structure having four planes;

[0034] FIG. 4C depicts a top view of a portion of one embodiment of a block of memory cells;

[0035] FIG. 4D depicts a cross sectional view of a portion of one embodiment of a block of memory cells;

[0036] FIG. 4E depicts a cross sectional view of a portion of one embodiment of a block of memory cells;

[0037] FIG. 4F is a cross sectional view of one embodiment of a vertical column of memory cells;

[0038] FIG. 4G is a schematic of a plurality of NAND strings in multiple regions of a same block;

[0039] FIG. 5 is a diagram depicting example threshold voltage distribution plot for a plurality of memory cells programmed to one bit per memory cell;

[0040] FIG. 6 is a schematic view of an example NAND string during a sensing operation;

[0041] FIG. 7 is a timing diagram for an example read operation of a 3D NAND memory;

[0042] FIG. 8A is a timing diagram for an example non-discharging read operation of a 3D NAND memory;

[0043] FIG. 8B is a timing diagram for another example non-discharging read operation of a 3D NAND memory;

[0044] FIG. 8C is a timing diagram for still another example non-discharging read operation of a 3D NAND memory;

[0045] FIG. 9 is a schematic view of an example computing system that includes a single processing unit and a plurality of high bandwidth flash packages; and

[0046] FIG. 10 is a flow chart depicting the steps of operating a memory device according to an example embodiment of the present disclosure.

## DETAILED DESCRIPTION OF THE ENABLING EMBODIMENTS

[0047] Technology is described for increasing the bandwidth and improving the power efficiency of NAND memory to provide a viable alternative to HBM devices, especially for computing systems that are optimized for large language model processing in artificial intelligence applications. In embodiments, technology is described that eliminates the repeated ramping up and down of unselected word lines between successive read operations. More specifically, a “non-discharging read” mode or technique is implemented in which a plurality of unselected word lines in a memory block remain biased at a read pass voltage

VREAD and are not discharged between successive read operations. By not discharging the unselected word lines, the time between read operations can be reduced, thereby improving read performance and read bandwidth.

**[0048]** FIG. 1 is a block diagram of one embodiment of a storage system **100** that implements the proposed technology described herein. In one embodiment, the storage system **100** is a solid state drive (“SSD”). The storage system **100** also can be a memory card, a USB drive, or any other type of storage system. In other words, the proposed technology is not limited to any one type of memory system.

**[0049]** The storage system **100** is connected to a host **102**, which can be a computer; server; electronic device (e.g., smart phone, tablet or other mobile device); appliance; or another apparatus that uses memory and has data processing capabilities. In some embodiments, the host **102** is separate from, but connected to, the storage system **100**. In other embodiments, the storage system **100** is embedded within the host **102**.

**[0050]** The components of the storage system **100** depicted in FIG. 1 are electrical circuits. The storage system **100** includes a memory controller **104** connected to non-volatile memory **106** and local high speed volatile memory **108** (e.g., DRAM). A local high speed volatile memory **108** is used by memory controller **104** to perform certain functions. For example, the local high speed volatile memory **108** stores logical to physical address translation tables (“L2P tables”).

**[0051]** The memory controller **104** includes a host interface **110** that is connected to and in communication with the host **102**. In one embodiment, a host interface **110** implements an NVM Express (NVMe) over PCI Express (PCIe). Other interfaces can also be used, such as SCSI, SATA, etc. The host interface **110** also is connected to a network-on-chip (NOC) **112**.

**[0052]** An NOC is a communication subsystem on an integrated circuit. The NOC’s can span synchronous and asynchronous clock domains or use un-clocked asynchronous logic. NOC technology applies networking theory and methods to on-chip communications and brings notable improvements over conventional bus and crossbar interconnections. The NOC improves the scalability of systems on a chip (SoC) and the power efficiency of complex SoCs compared to other designs.

**[0053]** The wires and the links of the NOC are shared by many signals. A high level of parallelism is achieved because all links in the NOC can operate simultaneously on different data packets. Therefore, as the complexity of integrated subsystems keep growing, a NOC provides enhanced performance (such as throughput) and scalability in comparison with previous communication architectures (e.g., dedicated point-to-point signal wires, shared buses, or segmented buses with bridges). In other embodiments, the NOC **112** can be replaced by a bus.

**[0054]** Connected to and in communication with NOC **112** is a processor **114**, an ECC engine **116**, a memory interface **118**, and a DRAM controller **120**. The DRAM controller **120** is used to operate and communicate with local high speed volatile memory **108** (e.g., DRAM). In other embodiments, the local high speed volatile memory **108** can be SRAM or another type of volatile memory.

**[0055]** In operation, the processor **114** performs the various controller memory operations, such as programming, erasing, reading, and memory management processes. In

one embodiment, the processor **114** is programmed by firmware. In other embodiments, the processor **114** is a custom and dedicated hardware circuit without any software. The processor **114** also implements a translation module, as a software/firmware process or as a dedicated hardware circuit.

**[0056]** In many systems, the non-volatile memory is addressed internally to the storage system using physical addresses associated with one or more memory dies. However, the host system will use logical addresses to address the various memory locations. This enables the host to assign data to consecutive logical addresses, while the storage system is free to store the data as it wishes among the locations of the one or more memory dies. To implement this system, the memory controller **104** (e.g., the translation module) performs address translation between the logical addresses used by the host and the physical addresses used by the memory dies.

**[0057]** One example implementation is to maintain tables (i.e., the L2P tables referenced above) that identify the current translation between logical addresses and physical addresses. An entry in the L2P table may include an identification of a logical address and corresponding physical address. Although logical address to physical address tables (or L2P tables) include the word “tables” they need not literally be tables. Rather, the logical address to physical address tables (or L2P tables) can be any type of data structure. In some examples, the memory space of a storage system is so large that the local memory **108** cannot hold all of the L2P tables. In such a case, the entire set of L2P tables are stored in non-volatile memory **106** and a subset of the L2P tables are cached (L2P cache) in the local high speed volatile memory **108**.

**[0058]** The ECC engine **116** performs error correction services. For example, the ECC engine **116** performs data encoding and decoding, as per an implemented ECC technique. In one embodiment, the ECC engine **116** is an electrical circuit programmed by software. For example, the ECC engine **116** can be a processor that can be programmed. In other embodiments, the ECC engine **116** is a custom and dedicated hardware circuit without any software. In another embodiment, the function of ECC engine **116** is implemented by the processor **114**.

**[0059]** The memory interface **118** communicates with the non-volatile memory **106**. In one embodiment, the memory interface provides a Toggle Mode interface. However, other interfaces also can be used. In some example implementations, the memory interface **118** (or another portion of the controller **104**) implements a scheduler and buffer for transmitting data to and receiving data from one or more memory die.

**[0060]** In one embodiment, the non-volatile memory **106** includes one or more memory die. FIG. 2A is a functional block diagrams of one embodiment of a memory die **200** that includes the non-volatile memory **106**. Each of the one or more memory dies of non-volatile memory **106** can be implemented as the memory die **200** of FIG. 2A. The components depicted in FIG. 2A are electrical circuits.

**[0061]** The memory die **200** includes a memory array **202** that can include non-volatile memory cells, as described in further detail below. The memory array **202** includes a plurality of layers of word lines that are organized as rows,

and a plurality of layers of bit lines that are organized as columns. However, other orientations can also be implemented.

[0062] The memory die 200 also includes row control circuitry 204, whose outputs 206 are connected to respective word lines of the memory array 202. In operation, the row control circuitry 204 receives a group of M row address signals and one or more various control signals from a system control logic circuit 208 and may include such circuits as row decoders 210, array terminal drivers 212, and block select circuitry 214 for both reading and writing (programming) operations.

[0063] The row control circuitry 204 also may include read/write circuitry. The memory die 200 also includes column control circuitry 216 including sense amplifier(s) 218 whose input/outputs 220 are connected to respective bit lines of the memory array 202. Although only a single block is shown for memory array 202, the memory die 200 can include multiple arrays that can be individually accessed.

[0064] The column control circuitry 216 receives a group of N column address signals and one or more various control signals from system control logic 208. The column control circuitry 216 may also include such circuits as column decoders 222; array terminal receivers or driver circuits 224; block select circuitry 226; read/write circuitry; and I/O multiplexers.

[0065] The system control logic 208 receives data and commands from memory controller 104 (FIG. 1) and provides output data and status to host 102. In some embodiments, the system control logic 208, which includes one or more electrical circuits, includes a state machine 228 that provides die-level control of memory operations. In one embodiment, the state machine 228 is programmable by software. In other embodiments, the state machine 228 does not use software and is completely implemented in hardware (e.g., electrical circuits). In another embodiment, the state machine 228 is replaced by a micro-controller or microprocessor, either on or off the memory chip.

[0066] The system control logic 208 also can include a power control module 230 that controls the power and voltages supplied to the rows and columns of memory structure 202 during memory operations and may include charge pumps and regulator circuits for creating regulating voltages. The system control logic 208 also includes storage 232 (e.g., RAM, registers, latches, etc.), which may be used to store parameters for operating memory array 202.

[0067] In operation, commands and data are transferred between the memory controller 104 and the memory die 200 via a memory controller interface 234 (also referred to as a “communication interface”). The memory controller interface 234 is an electrical interface for communicating with memory controller 104. Examples of the memory controller interface 234 include a Toggle Mode Interface and an Open NAND Flash Interface (ONFI). Other I/O interfaces can also be used in other embodiments.

[0068] In an embodiment, the system control logic 208 also includes column replacement control circuits 236, described in more detail below.

[0069] In some embodiments, all elements of the memory die 200, including the system control logic 208, can be formed as part of a single die. In other embodiments, some or all of the system control logic 208 can be formed on a different die.

[0070] In one embodiment, the memory structure 202 comprises a three-dimensional memory array of non-volatile memory cells in which multiple memory levels are formed above a single substrate, such as a wafer. The memory structure 202 may include any type of non-volatile memory that are monolithically formed in one or more physical levels of memory cells having an active area disposed above a silicon (or other type of) substrate. In one example, the non-volatile memory cells include charge-trapping layers and are arranged in a plurality of vertical NAND strings.

[0071] In another embodiment, the memory structure 202 includes a two-dimensional memory array of non-volatile memory cells. In one example, the non-volatile memory cells are NAND flash memory cells utilizing floating gates. Other types of memory cells (e.g., NOR-type flash memory) can also be used.

[0072] The exact type of memory array architecture or memory cell included in memory structure 202 is not limited to the examples above. Many different types of memory array architectures or memory technologies can be used to form memory structure 202. No particular non-volatile memory technology is required for purposes of the new claimed embodiments proposed herein. For example, suitable technologies for the memory cells of the memory structure 202 include ReRAM memories (resistive random access memories), magnetoresistive memory (e.g., MRAM, Spin Transfer Torque MRAM, Spin Orbit Torque MRAM), FeRAM, phase change memory (e.g., PCM), and the like. Examples of suitable technologies for memory cell architectures of memory structure 202 include two dimensional arrays, three dimensional arrays, cross-point arrays, stacked two dimensional arrays, vertical bit line arrays, and the like. One example of a ReRAM cross-point memory includes reversible resistance-switching elements arranged in cross-point arrays accessed by X lines and Y lines (e.g., word lines and bit lines).

[0073] In another embodiment, the memory cells may include conductive bridge memory elements. A conductive bridge memory element may also be referred to as a programmable metallization cell. A conductive bridge memory element may be used as a state change element based on the physical relocation of ions within a solid electrolyte. In some cases, a conductive bridge memory element may include two solid metal electrodes, one relatively inert (e.g., tungsten) and the other electrochemically active (e.g., silver or copper), with a thin film of the solid electrolyte between the two electrodes. As temperature increases, the mobility of the ions also increases causing the programming threshold for the conductive bridge memory cell to decrease. Thus, the conductive bridge memory element may have a wide range of programming thresholds over temperature.

[0074] Another example is magnetoresistive random access memory (MRAM) that stores data by magnetic storage elements. The elements are formed from two ferromagnetic layers, each of which can hold a magnetization, and the ferromagnetic layers are separated by a thin insulating layer. One of the two ferromagnetic layers is a permanent magnet that is set to a particular polarity, and the other ferromagnetic layer's magnetization can be changed to match that of an external field to store memory. The memory array may be built from a grid of such memory cells. In one embodiment, for programming, each memory cell lies between a pair of write lines that are arranged at right angles to each other, parallel to the cell, one above and one below

the cell. When current is passed through the write lines, an induced magnetic field is created. MRAM based memory embodiments will be discussed in more detail below.

**[0075]** Phase change memory (PCM) exploits the unique behavior of chalcogenide glass. One embodiment uses a GeTe—Sb<sub>2</sub>Te<sub>3</sub> super lattice to achieve non-thermal phase changes by simply changing the co-ordination state of the Germanium atoms with a laser pulse (or light pulse from another source). Therefore, the doses of programming are laser pulses. The memory cells can be inhibited by blocking the memory cells from receiving the light. In other PCM embodiments, the memory cells are programmed by current pulses. Note that the use of “pulse” in this document does not require a square pulse but includes a (continuous or non-continuous) vibration or burst of sound, current, voltage light, or another wave. These memory elements within the individual selectable memory cells, or bits, may include a further series element that is a selector, such as an ovonic threshold switch or metal insulator substrate.

**[0076]** The technology described herein is not limited to a single specific memory structure, memory construction or material composition, but covers many relevant memory structures within the spirit and scope of the technology as described herein and as understood by one of ordinary skill in the art.

**[0077]** The elements of FIG. 2A can be grouped into two parts: (1) the memory structure 202 and (2) peripheral circuitry, which includes all of the other components depicted in FIG. 2A. An important characteristic of a memory circuit is its capacity, which can be increased by increasing the area of the memory die of storage system 100 that is given over to the memory structure 202. However, this reduces the area of the memory die available for the peripheral circuitry. This can place quite severe restrictions on these elements of the peripheral circuitry. For example, the need to fit sense amplifier circuits within the available area can be a significant restriction on sense amplifier design architectures. With respect to system control logic 208, reduced availability of area can limit the available functions that can be implemented on-chip. Consequently, a basic trade-off in the design of a memory die for the storage system 100 may be the amount of area to devote to the memory structure 202 and the amount of area to devote to the peripheral circuitry.

**[0078]** Another area in which the memory structure 202 and the peripheral circuitry are often at odds is in the processing involved in forming these regions, since these regions often involve differing processing technologies and the trade-off in having differing technologies on a single die. For example, when the memory structure 202 is NAND flash, this is an NMOS structure, while the peripheral circuitry is often CMOS based.

**[0079]** Elements such as the sense amplifier circuits, charge pumps, logic elements in a state machine, and other peripheral circuitry in the system control logic 208 often employ PMOS devices. Processing operations for manufacturing a CMOS die will differ in many aspects from the processing operations optimized for an NMOS flash NAND memory or other memory cell technologies.

**[0080]** To improve upon these limitations, embodiments described below can separate the elements of FIG. 2A onto a separately formed die that is then bonded together with another die. More specifically, the memory structure 202 can be formed on one die (referred to as the memory die) and

some or all of the peripheral circuitry elements, including one or more control circuits, can be formed on a separate die (referred to as the control die). A memory die can be formed of just the memory elements, such as the array of memory cells of flash NAND memory, MRAM memory, PCM memory, ReRAM memory, or other memory type. Some or all of the peripheral circuitry, even including elements such as decoders and sense amplifiers, can then be moved on to a separate control die. This allows each of the memory die to be optimized individually according to its technology.

**[0081]** For example, a NAND memory die can be optimized for an NMOS based memory array structure, without worrying about the CMOS elements that have now been moved onto a control die that can be optimized for CMOS processing. This allows more space for the peripheral elements, which can now incorporate additional capabilities that could not be readily incorporated were they restricted to the margins of the same die holding the memory cell array.

**[0082]** The two die can then be bonded together in a bonded multi-die memory circuit, with the array on the one die connected to the periphery elements on the other die. Although the following will focus on a bonded memory circuit of one memory die and one control die, other embodiments can use more die, such as two memory die and one control die, for example.

**[0083]** FIG. 2B shows an alternative arrangement to that of FIG. 2A which may be implemented using wafer-to-wafer bonding to provide a bonded die pair. FIG. 2B depicts a functional block diagram of one embodiment of an integrated memory assembly 240. One or more integrated memory assemblies 240 may be used to implement the non-volatile memory 106 of storage system 100.

**[0084]** The integrated memory assembly 240 includes two types of semiconductor die (or more succinctly, “die”). The memory die 242 includes the memory structure 202 with the non-volatile memory cells. A control die 244 includes control circuitry 208, 216, and 204 (as described above). In some embodiments, the control die 244 is configured to connect to the memory structure 202 in the memory die 242. In some embodiments, the memory die 242 and control die 244 are bonded together.

**[0085]** FIG. 2B shows an example of the peripheral circuitry, including control circuits, formed in a peripheral circuit or control die 244 coupled to memory structure 202 formed in memory die 242. Common components are labelled similarly to FIG. 2A. The system control logic 208, the row control circuitry 204, and the column control circuitry 216 are located in the control die 244. In some embodiments, all or a portion of column control circuitry 216 and all or a portion of the row control circuitry 204 are located on memory die 242. In some embodiments, some of the circuitry in the system control logic 208 is located on the memory die 242.

**[0086]** The system control logic 208, the row control circuitry 204, and the column control circuitry 216 may be formed by a common process (e.g., CMOS process), so that adding elements and functions, such as the ECC controller, more typically found on a memory controller 104 may require few or no additional process steps, i.e., the same process steps used to fabricate controller 104 may also be used to fabricate the system control logic 208, the row control circuitry 204, and the column control circuitry 216.

**[0087]** Thus, while moving such circuits from a die such as the memory die 242 may reduce the number of steps

needed to fabricate such a die, adding such circuits to a die such as control die 244 may not require many additional process steps. The control die 244 also could be referred to as a CMOS die, due to the use of CMOS technology to implement some or all of the control circuitry 204, 208, 216.

[0088] FIG. 2B shows column control circuitry 216, including the sense amplifier(s) 218, on control die 244 coupled to memory structure 202 on memory die 242 through electrical paths 220. The electrical paths 220 may provide an electrical connection between the column decoder 222, the driver circuitry 224, the block select 226, and the bit lines of the memory structure 202. In an embodiment, the column control circuitry 216 also includes column replacement control circuits 236, which are described in more detail below.

[0089] Electrical paths may extend from the column control circuitry 216 in the control die 244 through pads on the control die 244 that are bonded to corresponding pads of the memory die 242, which are connected to the bit lines of the memory structure 202. Each bit line of the memory structure 202 may have a corresponding one of the electrical paths 220, including a pair of bond pads, which connects to the column control circuitry 216.

[0090] Similarly, the row control circuitry 204, including the row decoder 210, the array drivers 212, and the block select 214 are coupled to the memory structure 202 through electrical paths 206. Each of the electrical paths 206 may correspond to a data containing word line, a dummy word line, or a select gate line. Additional electrical paths may also be provided between control die 244 and memory die 242.

[0091] For purposes of this document, the phrases “a control circuit,” “control circuitry,” or “one or more control circuits” can include any one of or any combination of the memory controller 104; the state machine 228; all or a portion of the system control logic 208; all or a portion of row control circuitry 204; all or a portion of column control circuitry 216; a microcontroller; a microprocessor; and/or other similar functioned circuits.

[0092] The control circuit can include hardware only or a combination of hardware and software (including firmware). For example, one or more controllers programmed by firmware to perform the functions described herein is one example of a control circuit. A control circuit can include a processor, FGA, ASIC, integrated circuit, or other type of circuit.

[0093] In some embodiments, there is more than one control die 244 and more than one memory die 242 in an integrated memory assembly 240. In some embodiments, the integrated memory assembly 240 includes a stack of multiple control dies 244 and multiple memory dies 242.

[0094] FIG. 3A depicts a side view of an embodiment of an integrated memory assembly 300 stacked on a substrate 302 (e.g., a stack including control die 304 and memory die 306). In this embodiment, the integrated memory assembly 300 has three control die 304 and three memory die 306. In some embodiments, there are more than three memory die 306 and more than three control die 304.

[0095] Each control die 304 is affixed (e.g., bonded) to at least one memory die 306. Some of the bond pads 308/310 are depicted, although there may be many more bond pads. A space between two die 306, 304 that are bonded together is filled with a solid layer 312, which may be formed from epoxy or other resin or polymer. This solid layer 312 protects

the electrical connections between the die 306, 304 and further secures the die together. Various materials may be used as solid layer 312, but in some embodiments, it may be Hysol epoxy resin from Henkel Corp., having offices in California, USA.

[0096] Integrated memory assembly 300 may for example be stacked with a stepped offset, leaving the bond pads at each level uncovered and accessible from above. Wire bonds 314 connected to the bond pads connect control die 304 to substrate 302. A number of such wire bonds may be formed across the width of each control die 304 (i.e., into the page of FIG. 3A).

[0097] A memory die through silicon via (TSV) 316 may be used to route signals through each memory die 306. A control die TSV 318 may be used to route signals through each control die 304. The TSVs 316, 318 may be formed before, during or after formation of the integrated circuits in semiconductor die 306, 304. The TSVs may be formed by etching holes through the wafers. The holes may then be lined with a barrier against metal diffusion. The barrier layer may in turn be lined with a seed layer, and the seed layer may be plated with an electrical conductor such as copper, although other suitable materials such as aluminum, tin, nickel, gold, Solder balls 320 optionally may be affixed to contact pads 322 on a lower surface of substrate 302. Solder balls 320 may be used to couple integrated memory assembly 300 electrically and mechanically to a host device such as a printed circuit board. Solder balls 320 may be omitted where the integrated memory assembly 300 is to be used as an LGA package. Solder balls 320 may form a part of an interface between integrated memory assembly 300 and memory controller 104 (FIG. 1).

[0098] FIG. 3B depicts a side view of another embodiment of an integrated memory assembly 300 stacked on a substrate 302. The integrated memory assembly 300 of FIG. 3B has three control die 304 and three memory die 306. In some embodiments, there are many more than three memory die 306 and many more than three control die 304. In this example, each control die 304 is bonded to at least one memory die 306. Optionally, a control die 304 may be bonded to two or more memory die 306.

[0099] Some of the bond pads 308, 310 are depicted, but there may be many more bond pads than are illustrated. A space between two die 306, 304 that are bonded together is filled with a solid layer 312, which may be formed from epoxy or other resin or polymer. In contrast to the example in FIG. 3A, the integrated memory assembly 300 of FIG. 3B does not have a stepped offset. A memory die TSV 316 may be used to route signals through each memory die 306. A control die TSV 318 may be used to route signals through each control die 304.

[0100] As has been briefly discussed above, control die 304 and memory die 306 may be bonded together. Bond pads on each control die 304 and each memory die 306 may be used to bond the two die together. In some embodiments, the bond pads are bonded directly to each other, without solder or other added material, in a so-called Cu-to-Cu bonding process.

[0101] In a Cu-to-Cu bonding process, the bond pads are controlled to be highly planar and formed in a highly controlled environment largely devoid of ambient particulates that might otherwise settle on a bond pad and prevent a close bond. Under such properly controlled conditions, the

bond pads are aligned and pressed against each other to form a mutual bond based on surface tension.

**[0102]** As has been briefly discussed above, the control die **304** and the memory die **306** may be bonded together. Bond pads on each control die **304** and each memory die **306** may be used to bond the two die together. In some embodiments, the bond pads are bonded directly to each other, without solder or other added material, in a so-called Cu-to-Cu bonding process. In a Cu-to-Cu bonding process, the bond pads are controlled to be highly planar and formed in a highly controlled environment largely devoid of ambient particulates that might otherwise settle on a bond pad and prevent a close bond. Under such properly controlled conditions, the bond pads are aligned and pressed against each other to form a mutual bond based on surface tension. Such bonds may be formed at room temperature, though heat also may be applied. In embodiments using cu-to-cu bonding, the bond pads may be about 5  $\mu\text{m}$  square and spaced from each other with a pitch of 5  $\mu\text{m}$  to 5  $\mu\text{m}$ . Although this process is referred to herein as cu-to-cu bonding, this term also may apply even where the bond pads are formed of materials other than copper. When the area of bond pads is small, it may be difficult to bond the semiconductor die together. The size of and pitch between bond pads may be further reduced by providing a film layer on the surfaces of the semiconductor die including the bond pads. The film layer is provided around the bond pads. When the die are brought together, the bond pads may bond to each other, and the film layers on the respective die may bond to each other. Such a bonding technique may be referred to as hybrid bonding. In embodiments using hybrid bonding, the bond pads may be about 5  $\mu\text{m}$  square and spaced from each other with a pitch of 1  $\mu\text{m}$  to 5  $\mu\text{m}$ . Bonding techniques may be used providing bond pads with even smaller (or greater) sizes and pitches.

**[0103]** Some embodiments may include a film on a surface of the control die **304** and the memory die **306**. Where no such film is initially provided, a space between the die may be under filled with an epoxy or other resin or polymer. The under-fill material may be applied as a liquid which then hardens into a solid layer. This under-fill step protects the electrical connections between control die **304** and memory die **306**, and further secures the die together. Various materials may be used as under-fill material, such as Hysol epoxy resin from Henkel Corp., having offices in California, U.S. A.

**[0104]** FIG. 4A is a perspective view of a portion of one example embodiment of a monolithic three dimensional memory array/structure included in memory structure **202**, which includes a plurality non-volatile memory cells arranged as vertical NAND strings. For example, FIG. 4A shows a portion **400** of one block of memory. The structure depicted includes a set of bit lines BL positioned above a stack **402** of alternating dielectric layers and conductive layers. For example purposes, one of the dielectric layers is marked as D and one of the conductive layers (also called word line layers) is marked as W. The number of alternating dielectric layers and conductive layers can vary based on specific implementation requirements.

**[0105]** As will be explained below, in one embodiment the alternating dielectric layers and conductive layers are divided into, for example, four or five (or a different number of) regions by isolation regions IR. FIG. 4A shows one isolation region IR separating two regions. Below the alternating dielectric layers and word line layers is a common

source line layer SL. Memory holes are formed in the stack of alternating dielectric layers and conductive layers. For example, one of the memory holes is marked as MH. Note that in FIG. 4A, the dielectric layers are depicted as see-through so that the reader can see the memory holes positioned in the stack of alternating dielectric layers and conductive layers. In one embodiment, NAND strings are formed by filling the memory hole with materials including a charge-trapping material to create a vertical column of memory cells.

**[0106]** The non-volatile memory cells are arranged in memory holes, and each memory cell can store one or more bits of data, e.g., up to five bits of data per memory cell. More details of the three dimensional monolithic memory array that comprises memory structure **202** is provided below.

**[0107]** FIG. 4B is a block diagram explaining one example organization of memory structure **202**, which is divided into four planes **404**, **406**, **408** and **410**. Each plane is then divided into M blocks. In one example, each plane has about 2,000 blocks ("Block 0" to "Block M-1" with M being 2,000). However, different numbers of blocks and planes can also be used.

**[0108]** In one embodiment, a block of memory cells is a unit of erase. That is, all memory cells of a block are erased together. In other embodiments, the blocks can be divided into sub-blocks, each of which includes a plurality of word lines, and the sub-blocks can be the unit of erase. Memory cells also can be grouped into blocks for other reasons, such as to organize the memory structure to enable the signaling and selection circuits.

**[0109]** In some embodiments, a block represents groups of connected memory cells as the memory cells of a block share a common set of word lines. For example, the word lines for a block are all connected to all of the vertical NAND strings for that respective block. Although FIG. 4B shows four planes, each of which includes a plurality of blocks, more or fewer than four planes can be implemented in the memory structure **202**. In some embodiments, the memory structure includes eight planes.

**[0110]** Each block typically is divided into one or more pages, with each page being a unit of programming/writing and a unit of reading. Other units of programming also can be used. In an embodiment, one or more pages of data are typically stored in one row of memory cells. For example, one or more pages of data may be stored in memory cells connected to a common word line. In an embodiment, a page includes data stored in all memory cells connected to a common word line within the block.

**[0111]** FIGS. 4C-4G depict an example three dimensional ("3D") NAND structure that corresponds to the structure of FIG. 4A and can be used to implement the memory structure **202** of FIGS. 2A and 2B. FIG. 4C is a block diagram that depicts a top view of a portion **412** of Block **2** of plane **404**. As can be seen from FIG. 4C, the block depicted in FIG. 4C extends in the direction of **414**. In one embodiment, the memory array has many such layers with only the top layer being illustrated in FIG. 4C.

**[0112]** FIG. 4C depicts a plurality of circles that represent the memory holes, which are also referred to as vertical columns. Each of the memory holes/vertical columns includes multiple select transistors (also referred to as a select gate or selection gate) and multiple memory cells. In one embodiment, each memory hole/vertical column imple-

ments a NAND string. For example, FIG. 4C labels a subset of the memory holes/vertical columns/NAND strings **416**, **418**, **420**, **422**, **424**, **426**, **428**, **430**, and **432**.

[0113] FIG. 4C also depicts a set of bit lines **434**, including bit lines **436**, **438**, **440**, **442**, . . . **444**. FIG. 4C shows twenty four bit lines because only a portion of the block is depicted. It is contemplated that more than twenty four bit lines connected to memory holes/vertical columns of the block. Each of the circles representing memory holes/vertical columns has an “x” to indicate its connection to one of the bit lines. For example, bit line **436** is connected to the memory holes/vertical columns **418**, **420**, **422**, **426**, and **432**. The bit lines **436**, **438**, **440**, **442** also are in electrical communication with all other blocks in a given plane.

[0114] The block depicted in FIG. 4C includes a set of isolation regions **446**, **448**, **450** and **452**, which are formed of SiO<sub>2</sub>. However, other dielectric materials also can be used. Isolation regions **446**, **448**, **450**, and **452** serve to divide the top layers of the block into five regions or string groups. For example, the top layer depicted in FIG. 4C is divided into regions **454**, **456**, **458**, **460**, and **462**.

[0115] In one embodiment, the isolation regions only divide the layers used to implement select gates so that NAND strings in different regions/string groups (sometimes simply referred to as “strings”) can be independently selected. In one example implementation, a bit line connects to one memory hole/vertical column/NAND string in each of regions **454**, **456**, **458**, **460**, and **462**. In that implementation, each block has twenty-four rows of active columns and each bit line connects to five rows in each block.

[0116] In one embodiment, all of the five memory holes/vertical columns/NAND strings connected to a common bit line are connected to the same set of word lines; therefore, the system uses the drain side selection lines to choose one (or another subset) of the five to be subjected to a memory operation (program, verify, read, and/or erase).

[0117] FIG. 4C also shows Line Interconnects LI, which are metal connections to the source line SL from above the memory array. Line Interconnects LI are positioned adjacent regions **454** and **462**.

[0118] Although FIG. 4C shows each region **454**, **456**, **458**, **460**, and **462** as having four rows of memory holes/vertical columns, five regions and twenty four rows of memory holes/vertical columns in a block, those exact numbers are an example implementation. Other embodiments may include more or fewer regions per block; more or fewer rows of memory holes/vertical columns per region; and more or fewer rows of vertical columns per block.

[0119] FIG. 4C also shows the memory holes/vertical columns being staggered. In other embodiments, different patterns of staggering can be used. In some embodiments, the memory holes/vertical columns are not staggered.

[0120] FIG. 4D depicts a portion of one embodiment of a three dimensional memory structure **202** showing a cross-sectional view along line AA of FIG. 4C. This cross sectional view cuts through memory holes/vertical columns (NAND strings) **428** and **430** of region **462** (see FIG. 4C).

[0121] The structure of FIG. 4D includes two drain side select layers SGD0 and SGD1; two source side select layers SGS0 and SGS1; two drain side GIDL generation transistor layers SGDT0 and SGDT1; two source side GIDL generation transistor layers SGSB0 and SGSB1; two drain side dummy word line layers DD0 and DD1; two source side dummy word line layers DS0 and DS1; dummy word line

layers DU and DL that are separated by a joint; one hundred and sixty two word line layers WL0-WL161 for connecting to data memory cells; and dielectric layers DL. Other embodiments can implement more or fewer than the numbers described above for FIG. 4D. In one embodiment, SGD0 and SGD1 are connected together and SGS0 and SGS1 are connected together. In other embodiments, more or fewer SGDs (greater or lesser than two) are connected together and more or fewer SGS devices (greater or lesser than two) are connected together.

[0122] In one embodiment, erasing the memory cells is performed using gate induced drain leakage (GIDL), which includes generating charge carriers at the GIDL generation transistors such that the carriers get injected into the charge trapping layers of the NAND strings to change (reduce) respective threshold voltages V<sub>t</sub> of the memory cells. In the embodiment of FIG. 4D, there are two GIDL generation transistors at each end of the NAND string; however, in other embodiments there are more or fewer than two GIDL generation transistors.

[0123] Embodiments that use GIDL at both sides of the NAND string may have GIDL generation transistors at both sides. Embodiments that use GIDL at only the drain side of the NAND string may have GIDL generation transistors only at the drain side. Embodiments that use GIDL at only the source side of the NAND string may have GIDL generation transistors only at the source side.

[0124] The GIDL generation transistors have an abrupt PN junction to generate the charge carriers for GIDL and, during fabrication, a phosphorous diffusion is performed at the polysilicon channel of the GIDL generation transistors. In some cases, the GIDL generation transistor with the shallowest phosphorous diffusion is the GIDL generation transistor that generates the charge carriers during erase. However, in some embodiments charge carriers can be generated by GIDL at multiple GIDL generation transistors at a particular side of the NAND string.

[0125] The memory holes/vertical columns **428**, **430** are depicted protruding through the drain side select layers, source side select layers, dummy word line layers, GIDL generation transistor layers and word line layers. In one embodiment, each memory hole/vertical column comprises a vertical NAND string. Below the memory holes/vertical columns and the layers listed below is substrate **464**, an insulating film **466** on the substrate, and source line SL. The NAND string of memory hole/vertical column **428** has a source end at a bottom of the stack and a drain end at a top of the stack. As in agreement with FIG. 4C, FIG. 4D show vertical memory hole/column **428** connected to bit line **442** via connector **468**.

[0126] For ease of reference, drain side select layers, source side select layers, dummy word line layers, GIDL generation transistor layers and data word line layers collectively are referred to as conductive layers.

[0127] In one embodiment, the conductive layers are made from a combination of TiN and Tungsten. In other embodiments, other materials can be used to form the conductive layers, such as doped polysilicon, metal such as Tungsten, metal silicide, such as nickel silicide, tungsten silicide, aluminum silicide or the combination thereof.

[0128] In some embodiments, different conductive layers can be formed from different materials. Between conductive layers are dielectric layers DL. In one embodiment, the

dielectric layers are made from SiO<sub>2</sub>. In other embodiments, other dielectric materials can be used to form the dielectric layers.

[0129] The non-volatile memory cells are formed along memory holes/vertical columns which extend through alternating conductive and dielectric layers in the stack. In one embodiment, the memory cells are arranged in NAND strings. The word line layers WL0-WL161 connect to memory cells (also called data memory cells). The dummy word line layers connect to a plurality of dummy memory cells, which do not store data. In some embodiments, the data memory cells and the dummy memory cells may have a same structure. The drain side select layers SGD0 and SGD1 are used to electrically connect and disconnect the NAND strings to and from the bit lines. The source side select layers SGS0 and SGS1 are used to electrically connect and disconnect the NAND strings to and from the source line SL.

[0130] FIG. 4D shows that the memory array is implemented as a two tier architecture, with the tiers separated by a joint area. In one embodiment, it is expensive and/or challenging to etch so many word line layers intermixed with dielectric layers. To ease this burden, a first stack of word line layers (e.g., WL0-WL80) are laid down with alternating dielectric layers, then the Joint area is laid down, and next, a second stack of word line layers (e.g., WL81-WL161) are laid down with alternating dielectric layers. The joint area is thus positioned between the first stack of word line layers and the second stack of word line layers. In one embodiment, the joint areas are made from the same materials as the word line layers. In other embodiments, there can be no joint area or there can be multiple joint areas.

[0131] FIG. 4E depicts a portion of one embodiment of a three dimensional memory structure 202 showing a cross-sectional view along line BB of FIG. 4C. This cross sectional view cuts through memory holes/vertical columns (NAND strings) 416 and 470 of region 454 (see FIG. 4C). FIG. 4E shows the same alternating conductive and dielectric layers as FIG. 4D.

[0132] FIG. 4E also shows isolation region 446, which occupies a space that would have been used for a portion of the memory holes/vertical columns/NAND strings, including a space that would have been used for a portion of memory hole/vertical column 470. More specifically, a portion (e.g., half the diameter) of vertical column 470 has been removed in layers SGDT0, SGDT1, SGD0, and SGD1 to accommodate isolation region 446. Thus, while most of the vertical column 470 is cylindrical (has a circular cross section), the portion of vertical column 470 in layers SGDT0, SGDT1, SGD0, and SGD1 has a semi-circular cross section. In one embodiment, after the stack of alternating conductive and dielectric layers is formed, the stack is etched to create space for the isolation region and that space is then filled in with SiO<sub>2</sub>. This structure allows for separate control of SGDT0, SGDT1, SGD0, and SGD1 for regions 454, 456, 458, 460, and 462 (illustrated in FIG. 4C).

[0133] FIG. 4F depicts a cross sectional view of region 472 of FIG. 4D that includes a portion of memory hole/vertical column 428. In one embodiment, the memory holes/vertical columns are round. However, in other embodiments other shapes can be used. In one embodiment, memory hole/vertical column 428 includes an inner core layer 474 that is made of a dielectric, such as SiO<sub>2</sub>. Surrounding the inner core 474 is a polysilicon channel 476

(materials other than polysilicon can alternately be used). The channel 476 extends between and is connected with the bit line and the source line. Surrounding the channel 476 is a tunneling dielectric 478 layer, which may have an ONO structure. Surrounding the tunneling dielectric 478 layer is charge trapping layer 480, which may be formed of, for example, Silicon Nitride. It should be appreciated that the technology described herein is not limited to any particular material or structure.

[0134] FIG. 4F depicts the dielectric layers DL as well as the word line layers WL160, WL159, WL158, WL157, and WL156. Each of these word line layers includes a word line region 482 surrounded by an aluminum oxide layer 484, which is surrounded by a blocking oxide layer 486. In other embodiments, the blocking oxide layer 486 can be a vertical layer that is parallel with and adjacent to the charge trapping layer 480. The physical interaction of the word line layers with the vertical column forms the memory cells of the NAND string. Thus, in one embodiment a memory cell includes the channel 476, the tunneling dielectric 478, the charge trapping layer 480, the blocking oxide layer 486, the aluminum oxide layer 484, and the word line region 482. For example, word line layer WL160 and a portion of memory hole/vertical column 428 comprise a memory cell MC1. Word line layer WL159 and a portion of memory hole/vertical column 428 comprise a memory cell MC2. Word line layer WL158 and a portion of memory hole/vertical column 428 comprise a memory cell MC3. Word line layer WL157 and a portion of memory hole/vertical column 428 comprise a memory cell MC4. Word line layer WL156 and a portion of memory hole/vertical column 428 comprise a memory cell MC5. In other architectures, a memory cell may have a different structure; however, the memory cell would still be the storage unit.

[0135] When a memory cell is programmed, electrons are stored in a portion of the charge trapping layer 480 which is associated with (e.g., in) the memory cell. These electrons are drawn into the charge trapping layer 480 from the channel 476, through the tunneling dielectric 478, in response to an appropriate voltage on word line region 482. The threshold voltage (V<sub>th</sub>) of a memory cell is increased in proportion to the amount of stored charge.

[0136] In one embodiment, the programming is achieved through Fowler-Nordheim tunneling of the electrons into the charge trapping layer 480. During an erase operation, the electrons return to the channel 476 or holes are injected into the charge trapping layer 480 to recombine with electrons. In one embodiment, erasing is achieved using hole injection into the charge trapping layer 480 via a physical mechanism such as GIDL, as described above.

[0137] FIG. 4G is a schematic diagram of a portion of the three dimensional memory array depicted in in FIGS. 4B-4F. FIG. 4G shows physical data word lines WL0-WL161 running across the entire block. The structure of FIG. 4G corresponds to a portion 412 in Block 2 of FIG. 4B, including bit line 436. Within the block, in one embodiment, each bit line is connected to five NAND strings, one in each region of regions 454, 456, 458, 460, 462 (illustrated in FIG. 4C).

[0138] In one embodiment, the programming is achieved through Fowler-Nordheim tunneling of the electrons into the charge trapping layer 480. During an erase operation, the electrons return to the channel 476 or holes are injected into the charge trapping layer 480 to recombine with electrons. In

one embodiment, erasing is achieved using hole injection into the charge trapping layer 480 via a physical mechanism such as GIDL, as described above.

[0139] FIG. 4G is a schematic diagram of a portion of the three dimensional memory array depicted in in FIGS. 4B-4F. FIG. 4G shows physical data word lines WL0-WL161 running across the entire block. The structure of FIG. 4G corresponds to a portion 412 in Block 2 of FIG. 4B, including bit line 436. Within the block, in one embodiment, each bit line is connected to five NAND strings, one in each region of regions 454, 456, 458, 460, 462 (illustrated in FIG. 4C).

[0140] Similarly, the drain side select line/layer SGD1 is separated by isolation regions 446, 448, 450, and 452 (illustrated in FIG. 4C) to form SGD1-s0, SGD1-s1, SGD1-s2, SGD1-s3 and SGD1-s4 in order to separately connect to and independently control regions 454, 456, 458, 460, 462 (illustrated in FIG. 4C). The drain side GIDL generation transistor control line/layer SGDT0 is also separated by isolation regions 446, 448, 450 and 452 to form SGDT0-s0, SGDT0-s1, SGDT0-s2, SGDT0-s3 and SGDT0-s4 in order to separately connect to and independently control regions 454, 456, 458, 460, 462. Further, the drain side GIDL generation transistor control line/layer SGDT1 is separated by isolation regions 446, 448, 450 and 452 to form SGDT1-s0, SGDT1-s1, SGDT1-s2, SGDT1-s3 and SGDT1-s4 in order to separately connect to and independently control regions 454, 456, 458, 460, 462.

[0141] FIG. 4G only shows NAND strings connected to bit line 436. However, a full schematic of the block would show every bit line and five vertical NAND strings, which are in separate regions, connected to each bit line.

[0142] Although the example memories of FIGS. 4B-4G are three dimensional memory structures that include vertical NAND strings with charge-trapping material, other (2D and 3D) memory structures can also be used with the technology described herein.

[0143] The present disclosure is related to operating techniques that increase the bandwidth of a non-volatile memory device for the purpose of producing a high bandwidth flash (HBF) memory device that may be used as a replacement for HBM, particularly for machine learning inferencing operations. FIG. 9 illustrates an example embodiment of a computing system 900 constructed according to aspects of the present disclosure and that is optimized for large language model processing of machine learning interfacing operations. The computing system 900 includes a single graphics processor unit (GPU) 902 (or a similar processor unit) and eight HBF packages 904, which are all in electrical communication with the single GPU 902. Such a computing system 900 is particularly adapted for use in storing data pertaining to a large language mode because once the model data have been stored in the HBF packages 904, the model data are not updated or changed very often. Thus, for a machine learning inferencing application, the HBF packages 904 can be considered write a few times, read many times memory. Accordingly, read bandwidth is more important for the HBF packages 904 than write bandwidth. In some embodiments, the computing system 900 can include more or fewer than eight HBF packages 904. For example, in another embodiment, the computing system includes five HBF packages that are in electrical communication with a single GPU.

[0144] In an exemplary embodiment, each of the HBF packages 904 includes eight memory dies, each of which includes thirty-two planes that can be independently and simultaneously be operated on. Each plane includes a plurality of memory blocks with arrays of memory cells that are arranged in word lines and NAND strings. In some embodiments, the number of dies in each HBF packages and the number of planes per die can vary from these figures. In an example embodiment, in order to optimize the performance of the computing system 900 during large language model processing operations, the HBF packages 904 are able to communicate with the GPU 902 with a combined bandwidth during read of at least 3 TB/s.

[0145] The memory cells of the HBF packages 904 discussed above can be erased, programmed and read. At the end of a successful programming process, the threshold voltages  $V_t$  of the memory cells should be within one or more distributions of threshold voltages  $V_t$  for programmed memory cells or within a distribution of threshold voltages for erased memory cells, as appropriate. In other words, the threshold voltages  $V_t$  of the memory cells will fall within voltage ranges that are associated with different data states. FIG. 5 is a diagram of an example threshold voltage  $V_t$  distribution of a plurality of memory cells that are programmed to one bit of data per memory cell (single level cell or “SLC”). As illustrated, the memory cells fall into two different distributions or data states: erased (Er) and programmed (P). Memory cells that have threshold voltages  $V_t$  in threshold voltage distribution Er are in the erased data state, and memory cells that have threshold voltages  $V_t$  in the threshold voltage distribution P are in the programmed data state. In one embodiment, the erased memory cells store data “1” and the programmed memory cells store data “0.” FIG. 5 also depicts a read reference voltage  $V_{CGR}$  (sometimes also referred to as SLCR), which is set at a voltage between the two distributions. By testing (e.g., performing one or more sense operations) whether the threshold voltage  $V_t$  of a given memory cell is above or below  $V_{CGR}$ , the memory device can determine whether a memory cells is erased (data state Er) or programmed (data state P). FIG. 5 also depicts verify reference voltage  $V_v$ . In some embodiments, when programming memory cells to the programmed data state P, during the programming operation, the memory device will test whether those memory cells have a threshold voltage  $V_t$  greater than or equal to  $V_v$ . Programming continues for any given memory cell until its threshold voltage  $V_t$  exceeds the verify reference voltage  $V_v$ , at which point, programming ceases for that memory cell. In some cases, memory cells can be programmed to store more than one bit of data per memory cell in more than two data states. However, in order to optimize performance in the HBF packages of the subject disclosure, the SLC storage scheme is utilized in the exemplary embodiments.

[0146] In general, during read operations, the read reference voltage  $V_{CGR}$  is applied to a selected word line  $WL_n$ , which contains memory cells to be read, and the memory cells of the unselected word lines in the memory block are turned on with a read pass voltage  $V_{READ}$  (also referred to as a bypass voltage). In other words, the read pass voltage  $V_{READ}$  makes the memory cells of the unselected word lines conductive to electrons whether they are in the erased data state Er or the programmed data state P. FIG. 6 is a schematic view of an example NAND string and illustrating the voltages that are applied to the NAND string during the

sensing operation. A sense node is discharged through each NAND string that contains a memory cell to be read. Because the memory cells of the unselected word line are all turned on by the read pass voltage  $V_{READ}$ , a current through the NAND string is largely dictated by the threshold voltage  $V_t$  of the selected memory cell in the selected word line  $WLn$ . More specifically, if the current through the NAND string is relatively high, then the threshold voltage  $V_t$  of the selected memory cell is below the read reference voltage  $V_{CGR}$  and the selected memory cell is in the erased data state  $Er$ . On the other hand, if the current through the NAND string is relatively low, then the threshold voltage  $V_t$  of the selected memory cell is above the reference voltage  $V_{CGR}$  and the selected memory cell is in the programmed data state  $P$ .

[0147] As described in more detail below, the bandwidth of the HBF packages 904 (illustrated in FIG. 9) is increased by operating during sensing according to a “non-discharging read” mode of operation or sensing technique. The non-discharging read sensing technique improves bandwidth by reducing read latency during the sensing operations.

[0148] In a conventional 3D NAND memory, read operations to any particular block of memory cells occur randomly. 3D NAND memory is also susceptible to read disturb. One type of read disturb is caused by weak Fowler-Nordheim (F-N) tunneling due to a large voltage difference between the control gate and the channel of a cell. Another type of read disturb is hot electron injection (HEI) type of read disturb, which involves inadvertent programming of unselected memory cells adjacent to a selected word line. In particular, HEI read disturb may occur as a result of voltage gradients in the channels of the NAND strings of the unselected memory cells. The disturb results in a threshold voltage  $V_t$  upshift for a memory cell and is most pronounced for erased state memory cells. Referring back to FIG. 5, if the threshold voltages  $V_t$  of the memory cells that should be in the erased state  $Er$  rise above the reference voltage  $V_{CGR}$ , this can cause read errors.

[0149] FIG. 7 is a timing diagram for an example read operation of a selected memory cell. In particular, the diagram depicts example voltages of a selected word line  $WLn$ , unselected word lines  $WL_U$ , the control gate voltage of a selected SGD transistor  $SGD_S$ , the control gate voltage of the unselected SGD transistors  $SGD_U$ , and a bit line voltage coupled to a selected NAND string, which contains the selected memory cell, during the read operation.

[0150] FIG. 7 also depicts five different example phases  $\phi_1$ - $\phi_5$  during the read operation. During a first phase  $\phi_1$ , all word lines (the selected word line  $WLn$  and the unselected word lines  $WL_U$ ) are ramped up to the read pass voltage  $V_{READ}$ , and the control gate voltages of all SGD transistors (selected  $SGD_S$  and unselected  $SGD_U$ ) are ramped up to a high voltage  $V_H$  (e.g.,  $V_{DD}$ ) to turn ON (make conductive to electrons) the SGD transistors. This step is performed to equalize the potential in the channels of the NAND strings to reduce the impact of HEI read disturb.

[0151] Next, during a second phase  $\phi_2$ , the voltage on selected word line  $WLn$  is ramped down to a lower voltage (e.g., 0V), and the control gate voltages of all unselected SGD transistors  $SGD_U$  are ramped down to a low voltage  $V_L$  (e.g.,  $V_{SS}$ ) to turn OFF the unselected SGD transistors  $SGD_U$ .

[0152] During a third phase  $\phi_3$ , the voltage on selected word line  $WLn$  is ramped up to read reference voltage  $V_{CGR}$

and the bit line connected to the selected NAND string containing the selected word line is ramped up to a bit line voltage  $V_{BL}$ . After a time interval following the settling of these voltages settle, the actual read sensing occurs.

[0153] During a fourth phase  $\phi_4$ , the voltage on selected word line  $WLn$  is ramped up above read reference voltage  $V_{CGR}$ , and the voltages on unselected word lines  $WL_U$  and the control gate voltage of selected SGD transistor  $SGD_S$  begin ramping down to a very low voltage (e.g., 0V). The bit line voltage also begins ramping down to a low voltage (e.g., 0V).

[0154] During a fifth phase  $\phi_5$ , the voltage on selected word line  $WLn$ , the voltage on unselected word lines  $WL_U$  and the control gate voltage of selected SGD transistor  $SGD_S$  ramp down. The fourth phase  $\phi_4$  and the fifth phase  $\phi_5$  are for “channel clean” and ramp down to return the memory block back to a normal idle condition.

[0155] The timing diagram of FIG. 7, which involves ramping up and down the word lines and the control gates of the SGD transistors, is suitable for random reads of 3D NAND memory. As described above, however, a machine learning inferencing application has very intensive read requirements and the read operations are close to being sequential. Indeed, for a HBF application, the reads occur with very high frequency and because read performance is critical, there is significant motivation to reduce the time between successive read requests.

[0156] According to the non-discharging read techniques of the present disclosure, the repeated ramping up and down of the voltages is reduced to improve read performance by reducing the time delay between read operations. In particular, in an embodiment under certain conditions a “non-discharging read” technique is implemented in which the unselected word lines  $WL_U$  remain biased at the read pass voltage  $V_{READ}$  and do not ramp down between read operations.

[0157] In addition, according to the non-discharging read techniques, only the word lines and SGD transistors are ramped up and down depending on the relationship between successive read operations.

[0158] In an embodiment, read operations begin in accordance with a “normal read technique,” such as the technique depicted in FIG. 7 and described above. Then, if it is detected that there are more than a first predetermined number of reads to the same memory block  $N_{NDC}$  (e.g.,  $N_{NDC}=2$  or some other number) in less than a second predetermined time interval  $T_{NDC}$  (e.g.,  $T_{NDC}$ =several minutes at room temperature or several seconds at high temperature 85° C.), the non-discharging read technique is activated.

[0159] In an embodiment, the non-discharging read technique has three scenarios: (1) the previous read is on the same selected word line  $WLn$  but a different NAND string than the current read, (2) the previous read is on a different word line but the same NAND string as the current read, and (3) the previous read is on a different word line and a different NAND string than the current read. Example embodiments of each of these scenarios is described in turn.

[0160] FIG. 8A is a timing diagram for two successive read operations according to the non-discharging read techniques and in an HBF memory device (for example, one of the HBF packages 904 of FIG. 9). In particular, the diagram depicts the example voltages that are applied to a selected word line  $WLn$  (which in this case is  $WLn$ ), the unselected

word lines  $WL_U$ , the control gate voltage of a selected SGD transistor  $SGD_k$ , the control gate voltage of the unselected SGD transistors  $SGD_U$ , and a bit line voltage coupled to a selected NAND string during two successive read operations. Each of these two read operations includes three different example phases  $\phi_{2-x}$ - $\phi_{4-x}$ .

**[0161]** In the embodiment of FIG. 8A, the previous read operation (a first read operation) is on the same word line but a different NAND string than the current read operation (a second read operation). In particular, the previous read operation is on a first selected word line  $WLn$  and a first NAND string  $NS_k$ , and the current read operation is on the same first selected word line  $WLn$  but a second NAND string  $NS_{k+1}$ . During both of the previous read operation and the current read operation, all of the word lines other than first selected word line  $WLn$  are unselected word lines  $WL_U$ . In an embodiment, first NAND string  $NS_k$  includes a first SGD transistor  $SGD_k$ , and second NAND string  $NS_{k+1}$  includes a second SGD transistor  $SGD_{k+1}$ .

**[0162]** Throughout and between both read operations, and in the time between the two read operations, the voltage that is applied to the selected word line  $WLn$  (first selected word line  $WLn$ ) remains at the read reference voltage  $V_{CGR}$ , and the voltages that are applied to the unselected word lines  $WL_U$  remain at the read pass voltage  $V_{READ}$ . That is, the unselected word lines  $WL_U$  are not discharged between the first and second read operations.

**[0163]** During the first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ), the control gate of first SGD transistor  $SGD_k$  is biased to high voltage  $V_H$  to turn ON first SGD transistor  $SGD_k$ . Also during the first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ), the control gate of second SGD transistor  $SGD_{k+1}$  is biased to low voltage  $V_L$  to turn OFF second SGD transistor  $SGD_{k+1}$ . During phase  $\phi_{3-0}$  the bit line connected to first NAND string  $NS_k$  is ramped up to a bit line voltage  $V_{BL}$ . Some time interval after these voltages settle during phase  $\phi_{3-0}$ , the actual read sensing occurs of the selected memory cell at first selected word line  $WLn$  and first NAND string  $NS_k$ . As discussed above, the sensing process includes discharging a sense node through the NAND string  $NS_k$  and measuring the current through the NAND string  $NS_k$ .

**[0164]** During the second read operation (phases  $\phi_{2-1}$ - $\phi_{4-1}$ ), the control gate of first SGD transistor  $SGD_k$  ramps down to the low voltage  $V_L$  to turn OFF first SGD transistor  $SGD_k$ , and the control gate of second SGD transistor  $SGD_{k+1}$  is biased to the high voltage  $V_H$  to turn ON second SGD transistor  $SGD_{k+1}$ . During phase  $\phi_{3-1}$ , the bit line connected to second NAND string  $NS_{k+1}$  is ramped up to a bit line voltage  $V_{BL}$ . After these voltages settle during phase  $\phi_{3-1}$ , the actual read sensing occurs of the memory cell at first selected word line  $WLn$  and second NAND string  $NS_{k+1}$ . The sensing process includes discharging a sense node through the NAND string  $NS_{k+1}$  and measuring the current through the NAND string  $NS_{k+1}$ .

**[0165]** In an embodiment, during first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ) and second read operation (phases  $\phi_{2-1}$ - $\phi_{4-1}$ ) the control gates of all other unselected SGD transistors  $SGD_U$  remain at low voltage  $V_L$ . As depicted in dashed lines, during phases  $\phi_{2-0}$  and  $\phi_{2-1}$  the control gates of unselected SGD transistors  $SGD_U$  optionally may be ramped up to high voltage  $V_H$  and the down to low voltage  $V_L$  (e.g., for unselected SGD transistors  $SGD_U$  that have not been accessed in a long time).

**[0166]** FIG. 8B is a timing diagram for another example non-discharging read operation in a memory device of an HBF package. In particular, this diagram depicts example voltages of a selected word line  $WLs$  (which is  $WLn$  for the previous read operation and is  $WLn+1$  for the current read operation), unselected word lines  $WL_U$ , the control gate voltage of a selected SGD transistor  $SGD_k$ , the control gate voltage of unselected SGD transistors  $SGD_U$ , and a bit line voltage coupled to a selected NAND string during two successive read operations, each including three different example phases  $\phi_{2-x}$ - $\phi_{4-x}$  during each read operation.

**[0167]** In the example of FIG. 8B, the previous read operation (first read operation) is on a different word line but the same NAND string than the current read operation (second read operation). In particular, the previous read operation is on a first selected word line  $WLn$  and a first NAND string  $NS_k$ , and the current read operation is on a second selected word line  $WLn+1$  and the same first NAND string  $NS_k$ . During the previous read operation, all word lines other than first selected word line  $WLn$  are unselected word lines. During the current read operation all word lines other than second selected word line  $WLn+1$  are unselected word lines  $WL_U$ . In an embodiment, the first NAND string  $NS_k$  includes a first SGD transistor  $SGD_k$ .

**[0168]** Throughout both read operations, and in the time between the two read operations, the voltage of unselected word lines  $WL_U$  remains at read pass voltage  $V_{READ}$ . That is, unselected word lines  $WL_U$  are not discharged.

**[0169]** During the first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ), the first selected word line  $WLn$  is biased to the read reference voltage  $V_{CGR}$ , and second selected word line  $WLn+1$  is biased to the read pass voltage  $V_{READ}$ . The control gate of first SGD transistor  $SGD_k$  is biased to the high voltage  $V_H$  to turn ON the first SGD transistor  $SGD_k$ . During phase  $\phi_{3-0}$ , the bit line connected to first NAND string  $NS_k$  is ramped up to a bit line voltage  $V_{BL}$ . Some time interval after these voltages settle during phase  $\phi_{3-0}$ , the actual read sensing occurs of the memory cell at first selected word line  $WLn$  and first NAND string  $NS_k$ .

**[0170]** During the second read operation (phases  $\phi_{2-1}$ - $\phi_{4-1}$ ), the first selected word line  $WLn$  is biased to the read pass voltage  $V_{READ}$ , and second selected word line  $WLn+1$  is biased to the read reference voltage  $V_{CGR}$ . The control gate of first SGD transistor  $SGD_k$  is biased to low voltage  $V_L$  to turn OFF first SGD transistor  $SGD_k$ . During phase  $\phi_{3-1}$ , the bit line connected to the first NAND string  $NS_k$  is ramped up to a bit line voltage  $V_{BL}$ . Some time interval after these voltages settle during phase  $\phi_{3-1}$ , the actual read sensing occurs of the memory cell at second selected word line  $WLn+1$  and first NAND string  $NS_k$ .

**[0171]** In an embodiment, during first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ) and second read operation (phases  $\phi_{2-1}$ - $\phi_{4-1}$ ), the control gates of all other unselected SGD transistors  $SGD_U$  remain at low voltage  $V_L$ . As depicted in dashed lines, during phases  $\phi_{2-0}$  and  $\phi_{2-1}$ , the control gates of unselected SGD transistors  $SGD_U$  optionally may be ramped up to high voltage  $V_H$  and the down to low voltage  $V_L$  (e.g., for unselected SGD transistors  $SGD_U$  that have not been accessed in a long time).

**[0172]** FIG. 8C is a timing diagram for still another example non-discharging read operation in a memory device of an HBF package. In particular, the diagram depicts the example voltages of a selected word line  $WLs$  (which is  $WLn$  for the previous read operation and  $WLn+1$  for the

current read operation), unselected word lines  $WL_U$ , the control gate voltage of a selected SGD transistor  $SGD_k$ , the control gate voltage of a unselected SGD transistors  $SGD_U$ , and a bit line voltage coupled to a selected NAND string during two successive read operations, each including three different example phases  $\phi_{2-x}$ - $\phi_{4-x}$  during each read operation.

**[0173]** In the embodiment of FIG. 8C, the previous read operation (first read operation) is on both a different word line and a different NAND string than the current read operation (second read operation). In particular, the previous read operation is on a first selected word line  $WLn$  and a first NAND string  $NS_k$ , and the current read operation is on a second selected word line  $WLn+1$  and a second NAND string  $NS_{k+1}$ . During the previous read operation, all word lines other than first selected word line  $WLn$  are unselected word lines. During the current read operation, all word lines other than second selected word line  $WLn+1$  are unselected word lines. In an embodiment, first NAND string  $NS_k$  includes a first SGD transistor  $SGD_k$ , and second NAND string  $NS_{k+1}$  includes a second SGD transistor  $SGD_{k+1}$ .

**[0174]** Throughout both read operations, and in the time between the two read operations, the voltage of unselected word lines  $WL_U$  remains at read pass voltage  $V_{READ}$ . That is, unselected word lines  $WL_U$  are not discharged.

**[0175]** During the first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ) first selected word line  $WLn$  is biased to read reference voltage  $V_{CGR}$ , the second selected word line  $WLn+1$  is biased to the read pass voltage  $V_{READ}$ , the control gate of first SGD transistor  $SGD_k$  is biased to the high voltage  $V_H$  to turn ON first SGD transistor  $SGD_k$ , and the control gate of second SGD transistor  $SGD_{k+1}$  is biased to the low voltage  $V_L$  to turn OFF second SGD transistor  $SGD_{k+1}$ . During phase  $\phi_{3-0}$  the bit line connected to first NAND string  $NS_k$  is ramped up to a bit line voltage  $V_{BL}$ . Some time interval after these voltages settle during phase  $\phi_{3-0}$ , the actual read sensing occurs of the memory cell at first selected word line  $WLn$  and first NAND string  $NS_k$ .

**[0176]** During the second read operation (phases  $\phi_{2-1}$ - $\phi_{4-1}$ ) first word line  $WLn$  is biased to the read pass voltage  $V_{READ}$ , the second selected word line  $WLn+1$  is biased to the read reference voltage  $V_{CGR}$ , the control gate of first SGD transistor  $SGD_k$  is biased to the low voltage  $V_L$  to turn OFF first SGD transistor  $SGD_k$ , and the control gate of second SGD transistor  $SGD_{k+1}$  is biased to the high voltage  $V_H$  to turn ON second SGD transistor  $SGD_{k+1}$ . During phase  $\phi_{3-1}$  the bit line connected to second NAND string  $NS_{k+1}$  is ramped up to a bit line voltage  $V_{BL}$ . Some time interval after these voltages settle during phase  $\phi_{3-1}$ , the actual read sensing occurs of the memory cell at second selected word line  $WLn+1$  and second NAND string  $NS_{k+1}$ .

**[0177]** In an embodiment, during the first read operation (phases  $\phi_{2-0}$ - $\phi_{4-0}$ ) and the second read operation (phases  $\phi_{2-1}$ - $\phi_{4-1}$ ), the control gates of all other unselected SGD transistors  $SGD_U$  remain at the low voltage  $V_L$ . As depicted in dashed lines, during phases  $\phi_{2-0}$  and  $\phi_{2-1}$  the control gates of unselected SGD transistors  $SGD_U$  optionally may be ramped up to the high voltage  $V_H$  and the down to the low voltage  $V_L$  (e.g., for unselected SGD transistors  $SGD_U$  that have not been accessed in a long time).

**[0178]** Thus, as depicted in the example diagrams of FIGS. 9A-9B, throughout successive read operations, and in the time between two successive read operations, the voltage

of unselected word lines  $WL_U$  remains at read pass voltage  $V_{READ}$ . That is, the unselected word lines  $WL_U$  are not discharged.

**[0179]** By maintaining the voltage of unselected word lines  $WL_U$  at read pass voltage  $V_{READ}$  and not discharging the voltages of unselected word lines  $WL_U$ , read performance (read time  $t_R$ ) may be improved. By increasing the read performance (reducing read time  $t_R$ ), the bandwidth of the HBF memory device is increased.

**[0180]** FIG. 10 is a flow chart 1000 depicting the steps of performing a series of read operations according to an exemplary embodiment of the present disclosure. These steps could be performed by the controller; a processor or processing device or any other circuitry, executing instructions stored in memory; and/or other circuitry described herein that is specifically configured/programmed to execute the following steps.

**[0181]** At step 1002, a memory device, such as in one of the HBF packages 904 depicted in FIG. 9, is operating in a normal operating mode where the unselected word lines discharge between read operations. At step 1004, it is detected that a predetermined number of reads  $N_{NDC}$  has occurred in a predetermined time interval  $T_{NDC}$ . At step 1006, the memory device switches operation to the non-discharging read operating mode.

**[0182]** At step 1008, a first read operation is performed on a first memory cell with the unselected word lines being held at a read pass voltage  $V_{READ}$ . After the first read operation, the unselected word lines are not discharged from the read pass voltage  $V_{READ}$ . At step 1010, a second read operation is performed in a second memory cell. The second memory cell can be in the same word line as the first memory cell or a different word line than the first memory cell. Also, the second memory cell can be in the same string group as the first memory cell or a different string group than the first memory cell.

**[0183]** For purposes of this document, reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” or “another embodiment” may be used to describe different embodiments or the same embodiment.

**[0184]** For purposes of this document, a connection may be a direct connection or an indirect connection (e.g., via one or more other parts). In some cases, when an element is referred to as being connected or coupled to another element, the element may be directly connected to the other element or indirectly connected to the other element via intervening elements. When an element is referred to as being directly connected to another element, then there are no intervening elements between the element and the other element. Two devices are “in communication” if they are directly or indirectly connected so that they can communicate electronic signals between them.

**[0185]** For purposes of this document, the term “based on” may be read as “based at least in part on.”

**[0186]** For purposes of this document, without additional context, use of numerical terms such as a “first” object, a “second” object, and a “third” object may not imply an ordering of objects, but may instead be used for identification purposes to identify different objects.

**[0187]** For purposes of this document, the term “set” of objects may refer to a “set” of one or more of the objects.

**[0188]** The foregoing detailed description has been presented for purposes of illustration and description. It is not

intended to be exhaustive or to limit to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. The described embodiments were chosen in order to best explain the principles of the proposed technology and its practical application, to thereby enable others skilled in the art to best utilize it in various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the scope be defined by the claims appended hereto.

What is claimed is:

1. A memory device, comprising:
  - a memory block including a plurality of strings, each string comprising a plurality of non-volatile memory cells, each non-volatile memory cell being coupled to one of a plurality of word lines;
  - control circuitry coupled to the memory array, the control circuitry being configured to;
    - perform a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage, and
    - perform a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage; and
  - wherein the unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.
2. The memory device as set forth in claim 1, wherein the first non-volatile memory cell is in the same word line as the second non-volatile memory cell and is in a different string than the second non-volatile memory cell.
3. The memory device as set forth in claim 1, wherein the first non-volatile memory cell is in the same string as the second non-volatile memory cells and is in a different word line from the second non-volatile memory cell.
4. The memory device as set forth in claim 1, wherein the first non-volatile memory cell is in a different word line than and is in a different string than the second non-volatile memory cell.
5. The memory device as set forth in claim 1, wherein the first and second memory cells are programmed to one bit of data per memory cell.
6. The memory device as set forth in claim 1, wherein during the first read operation, the control circuitry discharges a current through the first non-volatile memory cell, and wherein during the second read operation, the control circuitry discharges a current through the second non-volatile memory cell.
7. A method of operating a memory device, comprising the steps of:
  - preparing a memory block that includes a plurality of strings, each string comprising a plurality of non-volatile memory cells, each non-volatile memory cell being coupled to one of a plurality of word lines;
  - performing a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage;
  - performing a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage; and

wherein the unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.

8. The method as set forth in claim 7, wherein the first non-volatile memory cell is in the same word line as the second non-volatile memory cell and is in a different string than the second non-volatile memory cell.

9. The method as set forth in claim 7, wherein the first non-volatile memory cell is in the same string as the second non-volatile memory cells and is in a different word line from the second non-volatile memory cell.

10. The method as set forth in claim 7, wherein the first non-volatile memory cell is in a different word line than and is in a different string than the second non-volatile memory cell.

11. The method as set forth in claim 7, wherein the first and second memory cells are programmed to one bit of data per memory cell.

12. The method as set forth in claim 7, wherein during the first read operation, the control circuitry discharges a current through the first non-volatile memory cell, and wherein during the second read operation, the control circuitry discharges a current through the second non-volatile memory cell.

13. A computing system, comprising:

a processing unit;

a plurality of high bandwidth flash packages in electrical communication with the processing unit, at least one of the high bandwidth flash packages including a memory block with a plurality of strings, each string comprising a plurality of non-volatile memory cells, each non-volatile memory cell being coupled to one of a plurality of word lines;

control circuitry coupled to the memory array, the control circuitry being configured to;

perform a first read operation on a first non-volatile memory cell in the memory block while a plurality of unselected word lines of the plurality of word lines are biased to a read pass voltage, and

perform a second read operation on a second non-volatile memory cell in the memory block while the plurality of unselected word lines are biased to the read pass voltage; and

wherein the unselected word lines are not discharged and remain biased to the read pass voltage during and between the first read operation and the second read operation.

14. The computing system as set forth in claim 13, wherein the first non-volatile memory cell is in the same word line as the second non-volatile memory cell and is in a different string than the second non-volatile memory cell.

15. The computing system as set forth in claim 13, wherein the first non-volatile memory cell is in the same string as the second non-volatile memory cells and is in a different word line from the second non-volatile memory cell.

16. The computing system as set forth in claim 13, wherein the first non-volatile memory cell is in a different word line than and is in a different string than the second non-volatile memory cell.

17. The computing system as set forth in claim 13, wherein the first and second memory cells are programmed to one bit of data per memory cell.

**18.** The computing system as set forth in claim **13**, wherein during the first read operation, the control circuitry discharges a current through the first non-volatile memory cell, and wherein during the second read operation, the control circuitry discharges a current through the second non-volatile memory cell.

**19.** The computing system as set forth in claim **13**, wherein the first and second read operations are large language model operations.

**20.** The computing system as set forth in claim **13**, wherein the plurality of high bandwidth flash packages are able to communicate with the processing unit during read operations at a bandwidth of at least 3 TB/s.

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